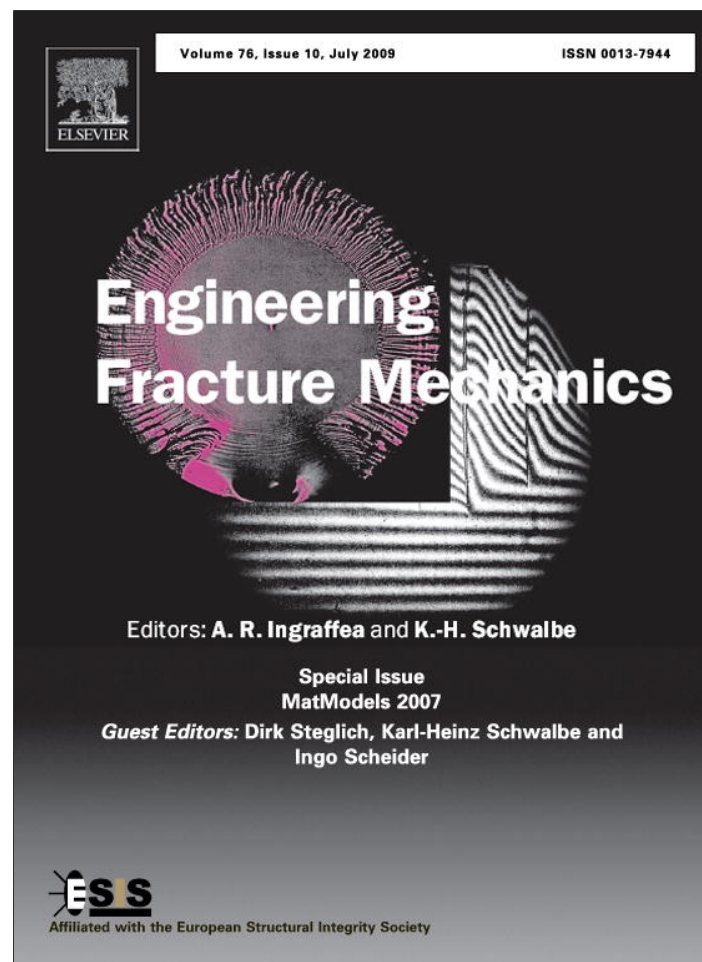




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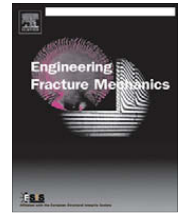
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DDSim: A hierarchical, probabilistic, multiscale damage and durability simulation system – Part I: Methodology and Level I

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ABSTRACT

Current tools for fatigue life prediction of metallic structural components are limited in some or all of the following capabilities: geometry of, and boundary conditions on, the affected structural component, automation of the simulation process, randomness of the primary variables, and physics of the damage evolution processes.

DDSim, a next-generation damage and durability simulator, addresses each of these limitations with a hierarchical, multiscale, “search and simulate” strategy. This hierarchical strategy consists of three levels. Level I, described in this paper, performs an initial, reduced order, conservative screening, based on a linear finite element analysis of the uncracked component, to determine the most life-limiting locations for intrinsic material flaws. Initial flaw size can be specified deterministically, or generated randomly from statistical descriptions of the microstructure and used in Monte Carlo simulation. The result is a scalar field of predicted life over the entire domain of the structure. The benefits of the Level I analysis include a high degree of automation, solution speed, and easy implementation of high performance parallel computing resources. A validation case study of Level I is described.

Levels II and III are outlined herein, but will be described in further detail in subsequent papers. The Level II analysis uses FRANC3D to accurately predict the number of cycles consumed by microstructurally large crack growth processes. Level III performs multiscale analyses to accurately predict the cycles consumed in microstructurally small crack growth processes.

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1. Introduction

Structural reliability is of utmost importance in structural design of space vehicles in extreme environments. Of particular interest to NASA's Project Constellation return-to-the-moon mission are highly reliable thermal protection systems (TPS) and primary airframe structure subjected to thermal gradients. Life and residual strength prediction under high reliability constraints is very difficult due to the lack of adequate models, experimental framework to guide design and manufacturing processes, and numerical simulation tools that accurately predict limit states. To that end, NASA's Constellation University Institutes Project (CUIP) [1] has formed a virtual institute, Structures and Materials for Extreme Environments (SMEE), whose focus is on research and development of tools and methodologies to support Project Constellation's mission.

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Nomenclature

RD, TD, ND	material rolling, transverse, normal direction
N^T	total fatigue life in cycles of loading (random variable)
$(\cdot)^{MLC}$	microstructurally large component of quantity $(\cdot)$
$(\cdot)^{MSC}$	microstructurally small component of quantity $(\cdot)$
$\oplus a^{MLC}$	summation that is dependent on the definition of a^{MLC}
X	random vector representing possible flaw location
A	random vector representing flaw geometry
E	random vector representing material resistance to crack growth
n^T	a sample of total fatigue life (deterministic)
n_{cr}^t	critical fatigue life
P_f	probability of failure
n_I	array containing components of life as computed by Level I
n_{II}	array containing components of life as computed by Level II
n_{III}	array containing components of life as computed by Level III
K_{max}	maximum mode one stress intensity factor in a fatigue loading program
K_{min}	minimum mode one stress intensity factor in a fatigue loading program
R	load ratio, K_{min}/K_{max}
K_{op}	opening stress intensity factor
K_c	critical stress intensity factor
ΔK	stress intensity factor range
ΔK_{eff}	effective stress intensity factor range
ΔK_{th}	threshold stress intensity factor range
da/dN	crack growth rate

Furthermore, the issue of aging aircraft is again receiving increasing attention recently from both our armed forces and the US Federal Aviation Administration. Specifically, the Defense Advanced Research Projects Agency (DARPA) has recognized the need for more reliable structural prognoses and, consequently, is currently funding an entire prognosis program [2]. Current life prediction tools and methods hinder flight/combat readiness due to imprecision and raise cost due to over, or misguided, conservatism. These undesirable effects are caused by models that are not physically grounded and under-value the inherently random nature of the material systems. The work reported herein addresses the needs identified in these NASA and DARPA projects.

Damage prognosis is the estimate of a structural system's remaining useful life [3]. There are many contributing elements to successful structural prognosis including quantifying the current status of the structural integrity, recording previous loading history, estimating future loading patterns and developing robust material models that account for such things as environmental effects. A next-generation tool is required to blend all of this useful, and generally random, information to simulate the structure's remaining life. Durability, on the other hand, is the structure's resilience to its loading conditions and its ability to maintain serviceability [4]. A durable structure can reduce inspection intervals, thereby reducing the cost of operation. A tool to simulate structural durability must incorporate much of the same input data as a damage prognosis tool and account for damage incubation and nucleation caused by defects within the material system itself.

The traditional damage tolerance approach to fatigue life assessment assumes defects – due to intrinsic material flaws, machining, or assembly processes – exist prior to service [5,6]. Damage tolerance procedures, by definition, do not account for fatigue life consumed by crack nucleation or crack growth prior to the assumed initial flaw size. This portion of the fatigue life of the vehicle can be substantial. The simplest and fastest damage tolerance assessment uses an approximate geometry and a fatigue analysis code such as NASGRO or AFGROW [7,8]. Software packages like these have been successfully designed to exploit the simplicity of hand-book stress intensity factor (SIF) solutions for fast calculations, but do so at the expense of fidelity. To a large degree, the accuracy of the predictions is dependent on the judgment of the engineer who must usually simplify geometry and local stress fields.

In the most elaborate procedure, finite element analysis (FEA) includes crack geometry in the structural model and allows the crack to evolve based on the stress field local to the crack front. The fields computed by the FEA are used in conjunction with a fracture mechanics code such as FRANC3D [9] to calculate the SIF history. The analyst must *manually* enter the computed SIF history into the fatigue analysis code to apply the fatigue crack growth model. If any probabilistic analysis is conducted, a third code, such as NESSUS [10], may be employed. NESSUS is a probabilistic driver that can be used in the damage tolerance assessment given information about random inputs. In the case of aircraft turbine rotor disks, a probabilistic life prediction code such as DARWIN [11] might be used in place the fatigue analysis code to perform probabilistic analysis.

The procedures of the current state-of-practice have been developed to take full advantage of currently available or readily measurable material data and, hence, use empirical crack growth rate models instead of models which simulate the physics of crack growth. The result is a practical and useful, yet limited, set of tools for fatigue life prediction. The least accurate

procedure is fast, but does not account for the arbitrary stress fields that can be present in complex geometry. The most accurate procedure is a time-consuming, multi-step, and multi-code effort.

In summary, the current technology of life prediction and damage tolerance tools exhibit some or all of the following deficiencies:

- Restricted physics – empirical crack growth laws; no multi-scale analysis; and averaged damage effects.
- Restricted flaw geometry – hand-book solutions; and absence of natural crack growth.
- Low automation – multiple codes requiring significant human intervention.
- No or little stochasticity – increased automation, unrestricted geometry and better physics would allow for the inherent stochasticity of the problem to be incorporated.

Clearly, there is high demand for a prognosis methodology that extends beyond current damage tolerance practice. An automated prognosis tool that embraces the physics and recognizes the probabilistic nature of the problem is highly desirable. To that end, this paper outlines a next generation Damage and Durability Simulator, DDSim, that has been developed using a hierarchical “search and simulate” strategy. Also presented is detailed information about implementation and validation of the first level of the hierarchy. DDSim Level I searches the domain of a structure using life prediction estimates based on current best-practice methods to choose sub-domains where the higher levels simulate the multiscale fatigue process with greater fidelity. The methodology presented herein is general enough to be extended to other material systems; however, the focus of this discussion will be on the aerospace-grade aluminum alloy 7075-T651.

The data demands for the next-generation methodology will require extensive materials databases, as will become evident in the subsequent section, and the computational demands will be extensive. However, experimental capabilities, data storage techniques and information sharing tools are undergoing commensurate technological advances. This methodology is driven by the notion that development of new tools should not be restricted by currently available technologies. Rather, new tools should be developed with an eye toward what will become available in the not-so-distant future.

The remainder of this paper is organized as follows: a description of the multiscale and stochastic nature of the fatigue damage in 7075-T651; an outline for DDSim, a new, three-tiered, methodology in damage and durability simulation; background for the Level I tool, the first of the three hierarchical tiers; implementation of the Level I tool; a validation case study of Level I; and conclusions.

2. The multiple length scales of fatigue crack growth in 7075-T651

Fatigue crack incubation in polycrystalline metals is inherently multiscale and random. Fig. 1(a) shows the inner-side of the bottom panel of an aircraft wing-skin, illustrating the *structural* length scale. Typically, fatigue cracking is localized near high stress concentration. This wing panel has a history of nucleating fatigue cracks along the row of bolt holes circled in the figure. Fig. 1(b) shows a detailed view of the geometry of one bolt hole that is approximately 5 mm deep. The bolt hole is counter-sunk for aerodynamic design purposes and counter-bored to reduce the stress concentration caused by the counter-sink. The geometry of, environment surrounding, boundary conditions on, and *microstructure* within the bolt hole combine to affect the origin of fatigue cracks.

Fig. 1(c) shows the fracture surface of the wing as a result of fatigue crack nucleation at the counter-sink, counter-bore intersection. Under periodic NDE, fatigue cracks down to 1 mm in surface length can be reliably discovered. However, up to 80% of the fatigue life of a structure can be expended in the processes leading up to a 1 mm crack [6,12]. Clearly, a thorough understanding of these processes is necessary to make reasonable predictions of total fatigue life.

Fig. 2(a) shows a micrograph of an approximate 0.001 mm³ volume of 7075-T651 [13]. The figure labels the material rolling direction (RD), transverse direction (TD), and normal direction (ND). An etching procedure was used to prepare the surfaces for the figure, exposing the crystalline structure. The various gray regions represent distinct grains. The constituent elements of 7075-T651 form micron-sized, second-phase precipitates – Al₁₂Fe₃Si, Al₆Cu, Al₆Fe, Al₇Cu₂Fe and Mg₂Si – which comprise approximately 2% of the bulk by volume. The iron-bearing precipitates, or particles, appear as dark black spots in the figure. These iron-bearing particles, which are very stiff and low toughness, play an all-critical role in the fatigue life of 7075-T651, as they are the primary source of fatigue crack nucleation.

Fig. 2(b) shows an approximate 1 mm² area of 7075-T651 exhibiting diffuse microcracking upon fatigue loading. In this figure, a different etching was used to prepare the surfaces. As a result, the second-phase particles appear light gray and the grain boundaries are much less distinct. Extensive microcracking like this is typical for situations where the peak load in the spectrum leads to localized yielding. At closer inspection, Fig. 2(b,inset), the microcracks obviously emanate from cracked particles. Indeed, Xue et al. have found that the primary fatigue crack incubation mechanism for 7075-T651 is broken iron-bearing particles [14]. A previous study by Merati and Eastaugh [15] was less conclusive; however, in a control set of experiments they also found broken particles to be the primary crack starters. Furthermore, while it has been observed that the Mg₂Si particles debond, they do not tend to spawn fatigue cracks in the surrounding matrix.

After rolling, the grains are flattened into pancaked shapes with average grain dimensions of 185 μm in the RD direction, 70 μm in the TD direction, and 18 μm in the ND direction [16,14]. The rolling process also affects the location of the second-phase by clustering the iron-bearing particles into stringers oriented in the RD direction. However, they are not preferentially located on grain boundaries [14]. At this length scale, the *microstructural*, the heterogeneity of the material is obvious. A

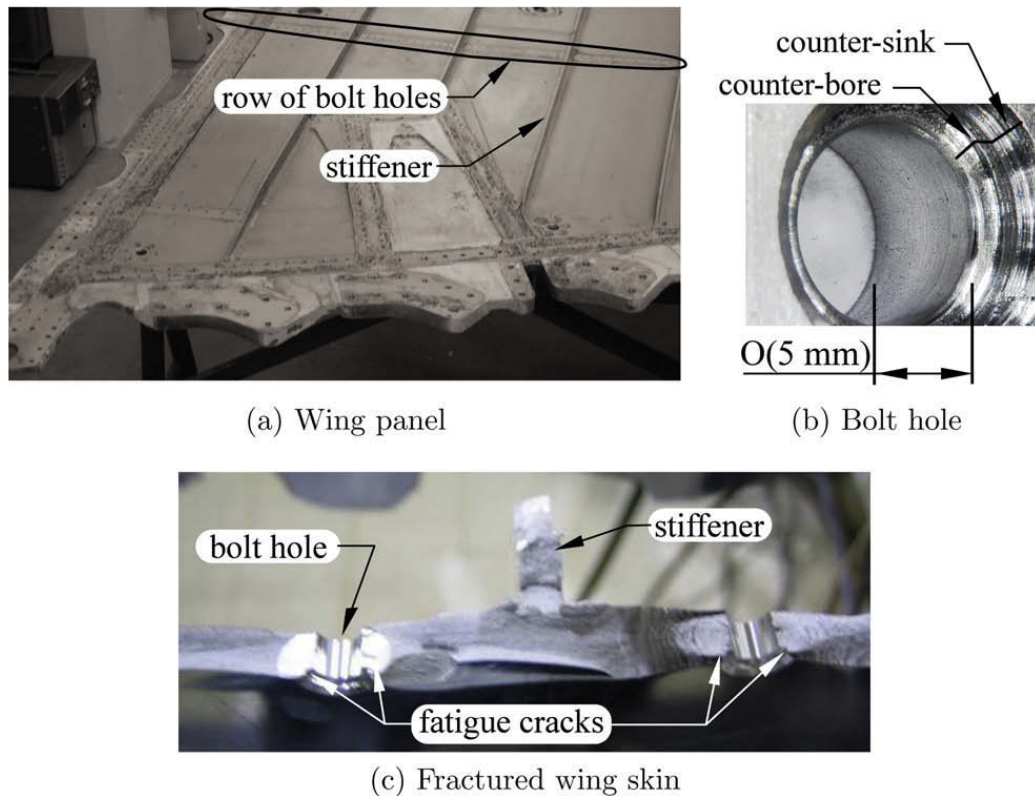


Fig. 1. (a) Inner-side of the bottom panel of an aircraft wing-skin. (b) A close-up view of a bolt hole. (c) Fracture surface at the line of bolt holes. (Photos courtesy of Northrop Grumman Corporation.).

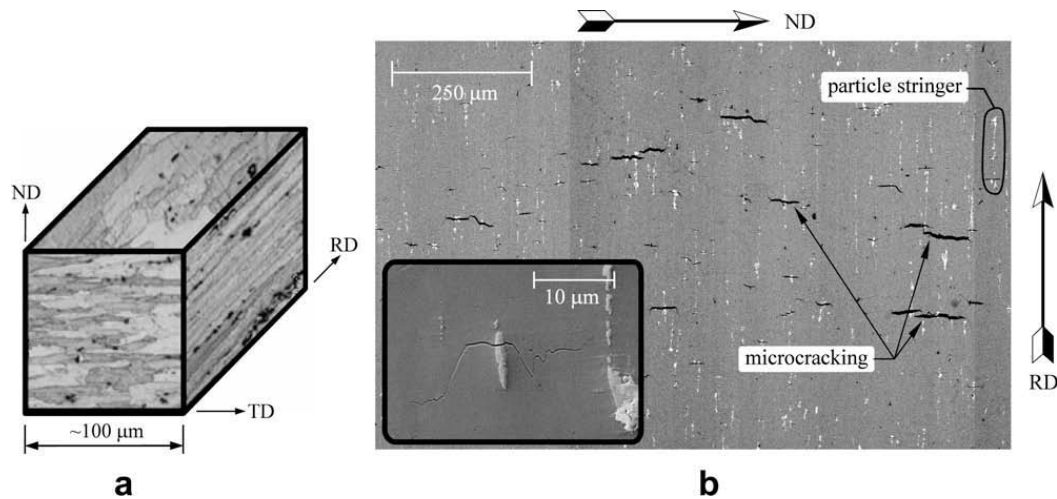


Fig. 2. (a) Micrograph displaying the microstructural morphology of 7075-T651 [13]. (b) Diffuse microcracking in a distinct two-phase material and fatigue crack nucleation at a broken second-phase particle (inset). (Photo courtesy of Northrop Grumman Corporation.).

single grain is transversely isotropic elastically, and highly anisotropic plastically, and the texture of the polycrystal is highly random. Obviously, even under uniform boundary conditions, the existence of a uniform stress field, at this length scale, is unlikely and impossible to predict without sufficient resolution and detail in the calculations.

The net result of the variability in microstructural geometry, texture and particle location causes fatigue crack growth to be highly random as seen in Fig. 3. This figure is the result of fatigue crack growth experiments conducted for micron-sized crack lengths [17] in AA 7075-T6. This figure shows that, under the same applied stress intensity, micron-sized crack growth rate can vary by an order of magnitude. For a given microcrack, originating from a cracked particle, the crack may find itself deep inside a grain or relatively near a grain boundary. Depending on the lattice orientation of the surrounding grain, the local stresses may be very high, or relatively low. The cracked particle may be one in a string of cracked particles which could result in crack shielding, or there may be an adjacent cracked particle with which to coalesce. The nearby grain boundary

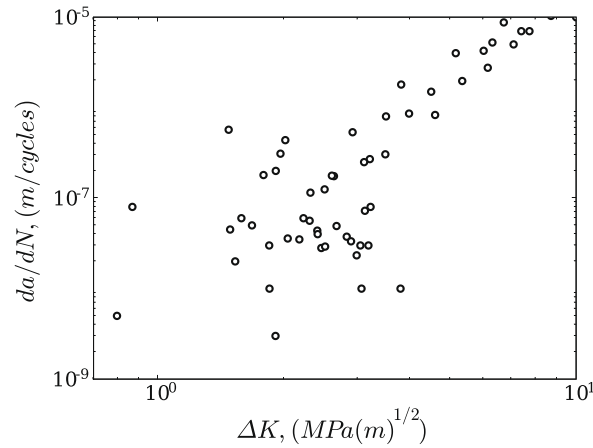


Fig. 3. Crack growth rate data in AA 7075-T6 showing highly random behavior [17].

may be the result of a highly misoriented neighboring grain, or a relatively shallow misorientation. It is clear that the majority of fatigue life consumed in microstructural crack growth processes is impossible to simulate without high fidelity, probabilistic models.

3. An outline for a new methodology: multiscale damage and durability simulation

The introduction to this paper argued for an improved methodology and tool for damage prognosis. This section outlines such a methodology and discusses the implementation of the methodology in DDSim. An integral part of life prediction is computing the fields that drive the damage mechanisms. DDSim is designed to refine this calculation, but only where it is most cost efficient. The crux of the problem is bridging the gap between the microstructural length scales – where damage nucleates and evolves – and the structural length scales used in modeling. The implicit challenge is to strike the most cost effective balance between computational supply and demand.

High resolution finite element models of the material microstructure can require millions of degrees of freedom (DOF) to model accurately cubic millimeters of volume [18,19]. Clearly, even with today's computing resources, modeling the microstructure throughout a human-serviceable engineering structure is not an option. DDSim addresses this problem by using multiple levels of fidelity of analysis, geometrical representation, computational mechanics, and material failure modeling. This hierarchical approach allows DDSim to link the length scales that matter with those of practical concern by focusing high performance (HP), parallel computing power on only the essential locations.

The methodology outlined in this paper is developed and described for fatigue cracking in 7075-T651. Every engineering material system comes with its own unique set of microstructural damage mechanisms, like those described for 7075-T651 in detail in Section 1. However, this methodology is directly extensible to a wide range of engineering metals, and analogously applicable to polymer matrix composites (PMCs). The remainder of this section provides the multiscale fatigue life prediction definitions used throughout the paper and describes the hierarchy of DDSim with a flowchart.

3.1. Definitions for a new methodology of fatigue damage and durability

DDSim addresses limitations of current simulators with a hierarchical “search and simulate” strategy consisting of three main levels. The strategy assumes that the total life of the structure can be decomposed as

$$N^T = N^{MLC} \oplus^{a^{MLC}} N^{MSC} \quad (1)$$

where N^T is the total fatigue life of the structure, N^{MLC} is the number of loading cycles consumed by microstructurally large crack (MLC) growth processes, N^{MSC} is the number of loading cycles consumed by microstructurally small crack (MSC) growth processes, and a^{MLC} is the characteristic length of a crack when it can be considered microstructurally large. The operator $\oplus^{a^{MLC}}$ implies summation that is dependent on the definition of a^{MLC} . This summation is highly dependent on the material system and often is the result of a committee decision. For this paper, a^{MLC} will be taken as 0.381 mm (0.015 in.), but it is understood that this size can be arbitrarily chosen by the engineer. The notation here will use capital letters to denote random variables and lower case letters to represent deterministic variables or specific samples of a random variable. Hence, while N^T is the random variable representing total fatigue life of the structure described by a probabilistic distribution, n^T is one sample of that total life and is given by an integer value of cycles.

In general, the functional dependence of N^T can be expressed by

$$N^T = N^T(\mathbf{X}, \mathbf{A}, \mathbf{E}, \mathbf{L}) \quad (2)$$

where $\mathbf{X} \in \mathfrak{R}^3$ is the dominant flaw location, $\mathbf{A}$, $\mathbf{E}$, and $\mathbf{L}$ are multi-dimensional arrays describing flaw geometry, material response including crack resistance and environmental effects, and the loading program, respectively. Because N^T is a function

of the random vectors $\mathbf{X}$, $\mathbf{A}$, $\mathbf{E}$ and $\mathbf{L}$, N^T is a random variable. Consequently, the probability of failure at a specified critical number of cycles can be defined as

$$P_f = P[N^T(\mathbf{X}, \mathbf{A}, \mathbf{E}, \mathbf{L}) \leq n_{cr}^T] \quad (3)$$

where n_{cr}^T is the critical number of loading cycles and the reliability of the structure is the probability of non-failure, or simply $1.0 - P_f$.

3.2. Flowchart for a new methodology of fatigue damage and durability

Fig. 4 shows a schematic of the data flow through DDSim to elucidate the strategy of the hierarchical architecture, and clearly delineate the three main levels. The components in the flowchart bear the following significance:

- dashed boxes indicate user input/control into the system;
- solid rectangular boxes indicate operations within the system (bold for primary hierarchical components);
- diamonds indicate decision points;
- ovals indicate probabilistic life predictions (output);
- arrows indicate the direction of data flow.

It should be emphasized that the decomposition indicated in Eq. 1 is *not* intended to suggest that N^T is evenly divided in two. In fact, as noted in Section 1, N^{MSC} can consume up to 80% of N^T [6,12]. In that case, it may not be cost effective to use Level II. Although it is not indicated in the flowchart, DDSim does not impose the use of Level II; the engineer can choose to skip directly to Level III. Conversely, for a vehicle already in service, the average microstructurally large crack length may be known by nondestructive evaluation. If so, the Level III simulation would be too costly and unnecessary, and the engineer would stop the simulation after Level II.

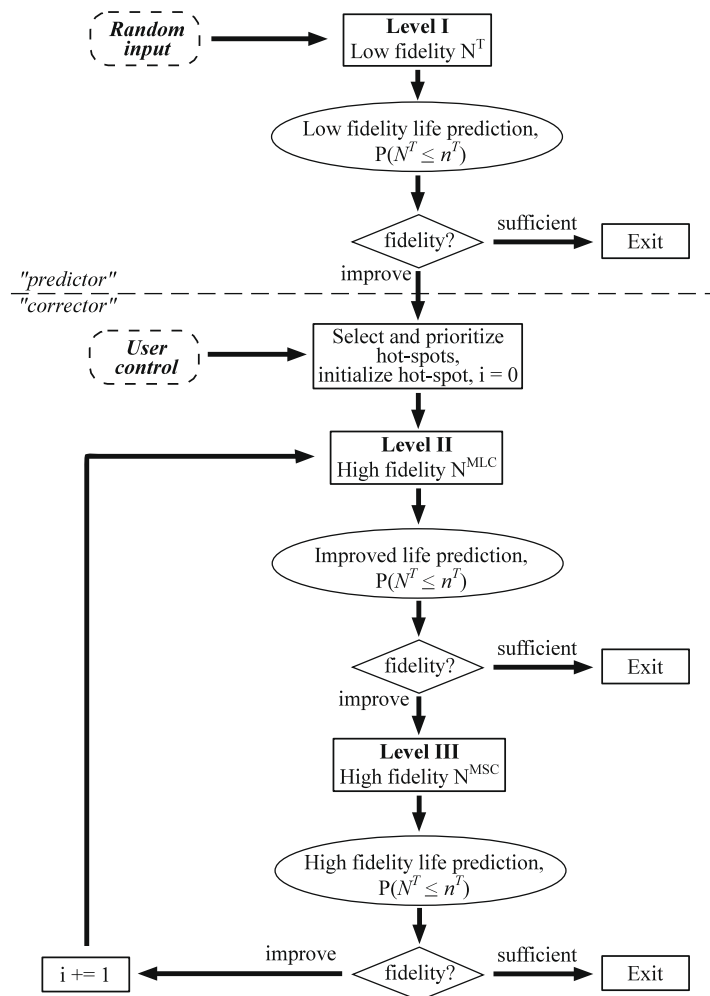


Fig. 4. Flowchart describing a new methodology for damage and durability simulation.

3.2.1. Random input

In general, “random input” includes boundary conditions, material properties, and microstructural geometry and texture. In this paper, random distributions will be used to describe microstructural geometry, microstructural texture, and dominant flaw location. However, the boundary conditions and material resistance to crack growth are currently considered deterministic. This changes the expression in Eq. (2) to

$$N^T = N^T(\mathbf{X}, \mathbf{A}). \quad (4)$$

The flowchart shows the input occurring once at the beginning of the simulation to facilitate discussion. However, it is not necessary that the implementation of the methodology follow this rule. Further discussion regarding input is included in the following subsections as appropriate.

3.2.2. Level I – reduced-order, probabilistic, low fidelity life prediction and initial screening

Level I acts as the “predictor” for the subsequent levels of analysis. Essentially, Level I is an automated way of performing state-of-practice, damage tolerance type assessments at every possible flaw location in a structure. Results from the Level I analysis are used in three ways: for selection of subdomains where the higher level simulations are performed; for a life prediction to be corrected by the higher level simulations; and for a life prediction for preliminary design and scoping. Level I idealizes the initial crack geometry as penny-shaped, semi-circular, or quarter-circular flaws for each node in the finite element mesh and uses *analytically computed* stress intensity factor (SIF) solutions.

For a deterministic analysis, the result is one estimate of total life at each finite element node. For a probabilistic analysis, the deterministic procedure is wrapped into Monte Carlo simulation and results in a set of samples from N^T for every finite element node, or for one node

$$\mathbf{n}_I^T = \{n_1^T, n_2^T, \dots, n_l^T\} \quad (5)$$

where the subscript I indicates that this is the Level I estimate, the n_j^T are the number of load cycles computed corresponding to initial crack size j , and l is the total number of initial crack sizes. From this set, a low fidelity estimate of total life, N^T , can be made. Further details of the Level I simulation are left to Sections 4 and 5. The salient features provided by the Level I life prediction are a high level of automation, probabilistic predictions, conservative calculation of SIFs and fast computations.

3.2.3. Fidelity

The fidelity of the predictions is to be assessed at the completion of each Level. The required fidelity is dictated by the engineer's confidence in the inputs, the required level of confidence in the reliability assessment, the overall accuracy of the calculations, and the desired level of reliability. Assessing the fidelity includes engineering judgment, calculating confidence intervals on the life predictions and assessing the accuracy of the individual models within each Level.

The issue of fidelity of life prediction can be thought of analogously to mesh refinement. Although there is not a rigorous proof of convergence, the blanket assertion is made here that fidelity in life prediction improves with improved representation of the physics driving fatigue damage. Hence, the assumptions in Level I which make it fast also make it reduced-order and result in a low fidelity life prediction. As such, the computed SIFs could be overestimated by as much as 50%, and corresponding approximations would be made for the estimate of N^{MSC} . To improve the fidelity, the engineer could refine the life prediction by using Level II to include better physical representation of the microstructurally large cracking processes at one, or all, of the finite element nodes. The Level II SIF calculations are much more accurate; however, the estimate of N^{MSC} is not improved. For another level of refinement, the engineer could use Level III to include the microstructural effects provided by the multiscale simulation at one, or all, of the finite element nodes. At this point, much of Level III is still basic research and an accurate estimate of the fidelity is an ongoing part of that effort.

To continue with the analogy, mesh refinement is performed in areas of high field gradients when higher resolution is required: too little refinement cannot produce accurate field calculations; too much leads to excessive cost. Similarly, there is an optimal amount of refinement in the life prediction that balances fidelity with cost. Performing the higher-order analyses of Levels II and III at all of the finite element nodes is not currently computationally feasible, nor physically justifiable.² The number of nodes to perform the higher order analyses is determined in the hot-spot selection stage, but directly affects the fidelity and cost. Higher-order analyses at more hot-spots means improved physics in more locations, but also more processing time.

If the fidelity of the final reliability estimate is insufficient, the engineer must refine the model. There are three ways the engineer can control the quality of the final estimate of reliability: provide higher resolution input data and run the entire system again; include a larger subset of the original domain to be analyzed in the corrector by selecting additional hot-spots; and include more or larger volume microstructural simulations.

² As of this writing in early-2009, it is reasonable to assume that an engineer would have computing power on the order of a GF/s readily available and usable on the desktop for DDSim, all Levels. It can be reasonably expected that this order will increase to a TF/s in special instances where there is access to a moderate sized parallel cluster, since DDSim has been developed with such an architecture in mind. It can also be reasonably expected that desktop order will increase generally to this level in the next few years. In this period, PF/s power will become available in special instances, and, at this level, full DDSim Level III coverage of a structural component becomes feasible. This is effectively unlimited computing power. What a wonderful world that will be. See for support of this prediction <http://www.top500.org/>.

3.2.4. Selection of hot-spots

Hot-spots are the locations where higher-order life predictions are to be made in the “corrector” phase of DDSim. The hot-spot locations are selected from the finite element node locations using a criterion based on the Level I life prediction. The user is required to define a criterion that suitably achieves the desired fidelity in simulation results. Any scalar, vector, or Voigt-form tensor field can be used, allowing the application of any number of user-defined criteria. For example, from a probabilistic Level I analysis, the user could compute the conditional probability of failure for a specified critical number of load cycles and use that for hot-spot selection. The resulting hot-spots would be a list of finite element node locations with a probability of failure within a specified bracketing interval. Two other criteria that can be used are: absolute lowest life, and mean life.

3.2.5. Level II – high fidelity calculation of N^{MLC}

DDSim Level II improves on the Level I life prediction by removing the restrictions that arise from using analytically computed SIF solutions. For each hot-spot, the Level II analysis uses FRANC3D [9] to automatically insert a crack of radius a^{MLC} into the geometry of the background FE model and to remesh appropriately. The boundary conditions are retained and the fields are recomputed using any available finite element code. Then the SIFs are extracted from the field calculations with FRANC3D. This process is repeated to generate a SIF history that is subsequently integrated to produce a high fidelity estimate of the number of cycles consumed by microstructurally large crack growth. Since the initial crack radius is given deterministically as a^{MLC} , the Level II estimate is deterministic. Hence, the notation that will be used for the Level II estimate is n_{II}^{MLC} .

If the number of cycles to grow the crack from its initial radius, a_j^0 , to a^{MLC} is recorded in Level I, a set similar to (5) can be written as

$$\mathbf{n}_I^{MSC} = \{n_1^{MSC}, n_2^{MSC}, \dots, n_l^{MSC}\} \quad (6)$$

where the n_j^{MSC} are the number of load cycles computed by Level I corresponding to initial crack size j , and l is total number of initial crack sizes. Next, employing the summation given in Eq. (1), $\mathbf{n}_I^{MSC}$ can be converted to the total fatigue life prediction from Level II by

$$\mathbf{n}_{II}^T = \{n_1^{MSC} + n_{II}^{MLC}, n_2^{MSC} + n_{II}^{MLC}, \dots, n_l^{MSC} + n_{II}^{MLC}\}. \quad (7)$$

Fig. 5 illustrates the scaling effect of n_{II}^{MLC} . It is clear from the figure that the Level II simulation does not change the shape of the conditional distribution of total fatigue life.

3.2.6. Level III – high fidelity calculation of N^{MSC}

DDSim Level III analysis provides an improvement on life prediction by performing coupled multiscale analyses which more accurately predict the number of cycles consumed by microstructurally small crack growth, N^{MSC} . The work to complete Level III is ongoing. This discussion describes the vision; however, it is missing some key details that are still under development.

The multiscale analysis incorporates detailed mesh models of the material microstructure. Current efforts include a “lumber” model which represents the geometry by stacking hexahedra whose dimensions are the average aspect ratio of observed grains and a more realistic representation of grain morphology generated by simulating the annealing process

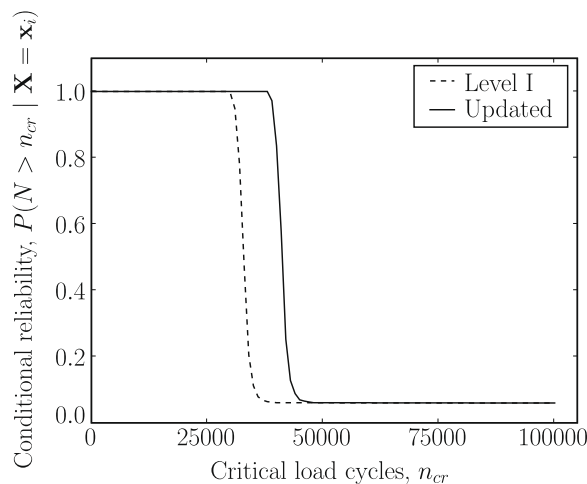


Fig. 5. n_{II}^{MLC} provides a scaling for the conditional reliability at node i .

[20,21]. The mechanics of the crystallographic deformation, including texture, are included with a crystal plasticity model [22].

There are two methods being developed to provide the coupling between lengths scales. For the sequel to this paper, a direct coupling is made using multipoint constraints. The second method uses a variation of the multi-grid method to transfer boundary conditions from the structural scale model to the microstructural model, again at the current hot-spot [23]. In either case, information flow is bi-directional: the boundary conditions applied to the structural scale model flow down-scale and the microstructural response is propagated up-scale.

With effectively unlimited computing power (see footnote above), Monte Carlo simulations could be conducted at each hot-spot. As such, many realizations of the microstructure would be generated and analyzed leading to a set of high fidelity predictions of the number of cycles consumed by microstructurally small crack growth processes

$$\mathbf{n}_{\text{III}}^{\text{MSC}} = \{n_1^{\text{MSC}}, n_2^{\text{MSC}}, \dots, n_l^{\text{MSC}}\}. \quad (8)$$

This, in turn, would be combined by Eq. 1 with $n_{\text{II}}^{\text{MLC}}$ for a set of high fidelity predictions of total life at each node.

4. Tools from the literature for the Level I life prediction

Level I idealizes initial crack geometry as penny-shaped, semi-circular, or quarter-circular and uses *analytical* stress intensity factor solutions to compute crack driving forces. In this section, we summarize the analytical stress intensity factor solutions, the fatigue crack growth rate equation, and the crack retardation model used to account for overloads.

4.1. Analytical and semi-analytical stress intensity factor solutions

The three analytical solutions for K_I encoded in DDSim Level I are: a fully-elliptical crack in an infinite body, and semi-elliptical and quarter-elliptical cracks in a finite-thickness plate.

The solution used in Level I for the full ellipse is derived by Roy and Saha [24] using the weight function method. It has been chosen for its ease-of-use and because it accounts for bi-quadratic stress fields on the crack surface. A three-dimensional weight function for arbitrary loading and geometry would be ideal for the Level I simulation; however, to the best of the authors' knowledge there is no such analytical solution available.

There are two approximate semi-analytical solutions, developed by Raju and Newman [25–29], encoded in Level I for surface cracks. One solution gives the stress intensity factor for a semi-elliptical surface crack in a finite-dimensioned plate subjected to remote tension and bending. The other solution gives the stress intensity factor for a quarter-elliptical corner crack in a finite-dimensioned plate subjected to remote tension and bending. These solutions are not based on the weight function method. Rather, they were developed by performing a large number of finite element analyses to cover various ranges of crack and plate geometries. The stress intensity factor solutions are given as equations fit to match the finite element simulations to within $\pm 5\%$ for most of the geometries.

For the purposes of Level I, the dimensions of the plate are assumed to be much larger than the crack and the Raju and Newman's solutions are applied to arbitrary geometry. Fig. 6 shows a comparison between the analytical SIF solutions and SIFs calculated using FRANC3D for varying crack radii. The applied loading is combined tension and bending. Fig. 6(a) shows the ratio of the SIFs for a semi-circular surface flaw and (b) shows the ratio for quarter-circular surface flaw. The conservative estimate at the interior point of the semi-circular crack and at the through thickness surface point on the quarter-circular

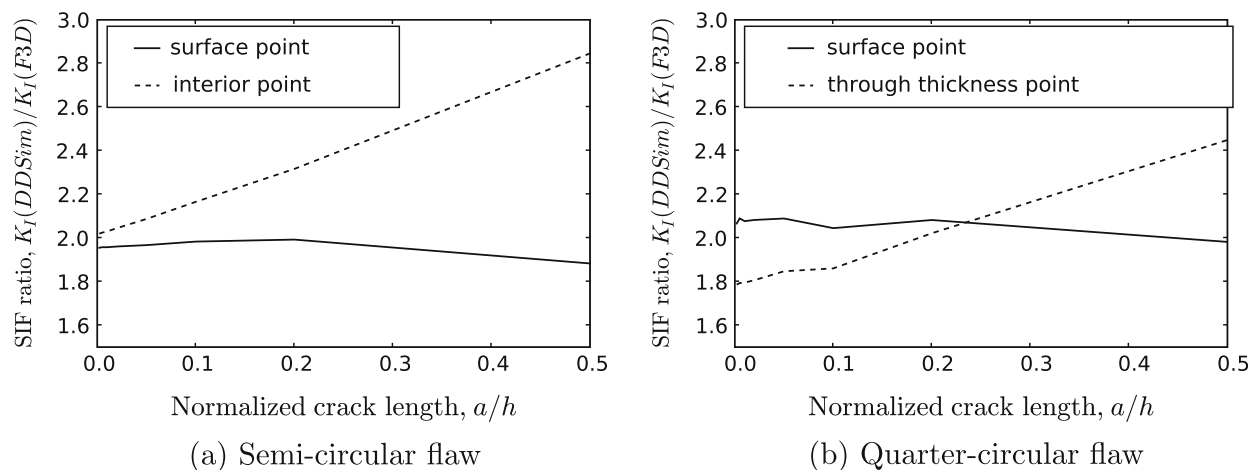


Fig. 6. Comparison of stress intensity factors. The ratios between DDSim's calculated SIFs and those calculate using FRANC3D are plotted versus crack lengths. The crack length is normalized by the plate thickness.

crack increases with crack size. This is due to the assumed infinite dimensions. It can be seen that the assumptions made in DDSim lead to generally conservative SIF estimates for this geometry.

4.2. Fatigue crack growth rate

The fracture mechanics approach to fatigue life prediction has traditionally used empirically derived models for crack growth rate. This section briefly presents the crack growth rate relationship used in Level I. Then, a model to account for the near-threshold behavior of young cracks is discussed. Finally, a crack retardation model from the literature has been implemented and is discussed.

4.2.1. The NASGRO rate equation

Crack growth rate models have evolved from the power-law form of the Paris model [30] to include terms that account for crack-closure and rate acceleration near the critical stress intensity. Crack-closure is the result of crack surface contact and other mechanisms prior to full unloading and is thought to cause, or be a main contributor to, the threshold behavior [31–33]. The consequence of crack-closure is that the minimum stress intensity factor at the crack front no longer corresponds with the minimum applied load in the loading program. Rather, the minimum stress intensity factor corresponds to the stress intensity required to open the crack, K_{op} , or the opening stress intensity. Thus, the stress intensity factor range is reduced and can be stated as

$$\Delta K_{eff} = K_{max} - K_{op}. \quad (9)$$

Newman [34] used a 2-D closure model to simulate the effects of plasticity-induced crack-closure. The result of that study is an equation that gives the crack-opening stress as a function of the maximum applied stress, the ratio of the applied stresses, and an empirical constraint factor which accounted for specimen thickness. Fig. 7 shows Newman's crack-opening stress intensity relation by plotting the ratio between opening stress intensity and maximum applied stress intensity, f , versus the load ratio, $R = \frac{K_{min}}{K_{max}}$. For $R \geq 0.7$, the ratios are approximately equal, which indicates that the crack-closure effect dissipates with increased minimum load.

Forman and Mettu [35] gave a crack growth rate relationship which more accurately follows the sigmoidal trend of fatigue data and is considered to be an excellent approximation for engineering application [12]. Their model is included in NASGRO (version 4.2) [7], referred to here as the NASGRO equation, and is the model used in Level I. Their crack growth rate equation is

$$\frac{da}{dN} = C(\Delta K_{eff})^n \frac{\left(1 - \frac{\Delta K_{th}}{\Delta K}\right)^p}{\left(1 - \frac{K_{max}}{K_c}\right)^q} \quad (10)$$

where

$$\Delta K_{eff} = \Delta K \left(\frac{1-f}{1-R} \right) \quad (11)$$

and $\Delta K = K_{max} - K_{min}$, f and R are as previously defined, ΔK_{th} is the threshold stress intensity range which is internally computed by empirical equations, K_c is an appropriate measure of fracture toughness, and C , n , p , and q are material dependent, empirical constants. Note that upon substitution for f and R in Eq. 11, Eq. 9 is recovered. The equations for Newman's f parameter and ΔK_{th} involve several more material parameters. These equations are not included here, but can be found in Refs. [34,7].

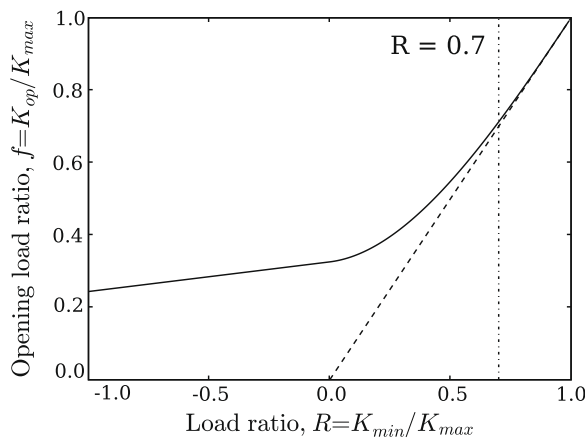


Fig. 7. Newman's crack-opening stress intensity relation.

4.2.2. Small cracks and a modification to the NASGRO equation

Small cracks in aluminum alloys exhibit anomalous behavior at near threshold stress intensity ranges [16,32,36]. First, for the same stress intensity range, small cracks generally grow at faster rates than large cracks [31]. Second, growth rates for small cracks show a great deal more variability than large cracks. Large cracks are generally considered to have minimum dimensions on the order of 2 mm [37]. The faster growth rate is often attributed to small cracks lacking plastic history. That is, small cracks are not inhibited by plasticity-induced crack-closure nor do they have a region of plastically-deformed material confining their crack fronts. From this point of view, it may be more appropriate to call them “young” cracks because lack of plastic history is more related to age than to crack size. However, the variability of the growth rates is primarily a result of the randomness of the microstructure [38], an essential observation for this work. Small crack growth is dominated by microstructural features such as particle inclusions, grain boundaries, grain orientation and grain morphology.

Eq. 10 does not, for the purposes of the Level I analysis, adequately account for the anomalous behavior of small cracks. Since it is important in Level I to remain conservative, faster growth rates of smaller cracks must be taken into account. To be sure, using linear elastic fracture mechanics to model fatigue crack growth at such small length scales is dubious. The maximum plastic zone for 7075-T651 is estimated to be on the order of a millimeter, whereas, these surface cracks are on the order of a tenth of a millimeter in size. However, given the scope of the Level I simulation, this approximation is accepted because Level III will accurately simulate the damage processes within the process zone.

McEvily and Ishihara [39] presented an elastic–plastic fatigue crack growth rate model for steel. In their model, they modified the SIF to account for crack tip blunting by treating the crack as a fine notch and adding its radius, a material constant, to the actual crack length. Furthermore, to account for crack-closure, or lack thereof, they introduced an effective opening stress intensity factor range

$$\Delta K_{op} = (1 - e^{-ka})(K_{op} - K_{min}) \quad (12)$$

where k is a measured material constant corresponding to the rate of development of crack-closure, a is the newly formed crack length, and K_{op} and K_{min} are as previously defined. Eq. 12 is then used to replace Eq. 11 so that

$$\Delta K_{eff} = \Delta K - \Delta K_{op} = K_{max} - K_{op} + e^{-ka}(K_{op} - K_{min}). \quad (13)$$

From this, it can be seen that for small crack dimensions, the stress intensity range reverts back to $\Delta K = K_{max} - K_{min}$. Conversely, for large crack dimensions, it takes the form of the original definition of the effective stress intensity factor range, shown in Eq. 9. This model removes plasticity-induced closure for small cracks as desired. However, as previously pointed out, the accumulation of plasticity is not as much a function of crack size as it is a function of crack age.

To address this problem, a new model is derived specifically for Level I that removes the closure effects for young cracks, irrespective of their size. Consequently, an effective load ratio is defined as

$$R_{eff} = (1 - e^{-N/damp})(R - 0.7) + 0.7 \quad (14)$$

where N is the current number of load cycles, $damp$ is a tuning parameter, and R is as previously defined. The parameter $damp$ controls how soon the R_{eff} returns to the applied R . Smaller values of $damp$ corresponds to a rapid return, whereas larger values corresponds to a slower return. With the effective load ratio defined, the NASGRO crack growth rate equation is modified such that

$$\frac{da}{dN} = \begin{cases} C_n(\Delta K)^{n'} & \Delta K \leq \Delta K_{th} \text{ and } 0.65 \leq R_{eff} \leq 0.7 \\ C(\Delta K_{eff})^n \left(\frac{1 - \frac{\Delta K_{th}}{\Delta K}}{1 - \frac{K_{max}}{K_c}} \right)^p & \text{otherwise.} \end{cases} \quad (15)$$

where n' is a user specified parameter, C_n is calculated internally to provide a smooth transition back to the NASGRO equation, and all other parameters are the same as for Eq. 10. R_{eff} is used in place of R to compute the crack growth rate *except* when computing the maximum stress intensity in the denominator of Eqs. 10 and 15.

The effective load ratio increases the stress intensity range for young cracks. This can be seen by examination of Eq. 11 and comparing Figs. 7 and 8. Fig. 8 shows the decay of R_{eff} with increasing load cycle. For young cracks, R_{eff} is larger than the actual applied load ratio which results in a smaller opening stress ratio and, thus, a larger effective stress intensity range.

Fig. 10 shows a set of crack growth rate data [17], the crack growth rate predicted using Eq. 10, and the crack growth rate predicted using Eq. 15. The material properties used to generate the curves are shown in Table 1 and are taken from the NASGRO material database. The crack growth data is generated from AA 7075-T6 single edge-notch tension specimens with initial flaw size 6 μm . AA 7075-T6 is very similar to 7075-T651, the alloy of primary interest in this paper. The predictions are based on the SIF solution, given in [17], for a semi-elliptical surface crack at the center of the edge-notch. The figure shows that the modified NASGRO equation, Eq. 15, has the desired effect of accelerating the small crack growth rate for this alloy when $damp = 1000.0$. This parameter is determined by trial and error to predict the higher growth rates observed, as seen in the figure.

Furthermore, young cracks exhibit higher growth rates at increasing stress levels as shown in Fig. 9, taken from Ref. [36]. In addition to trying to predict the higher observed rates, the $damp$ parameter is adjusted to simultaneously predict this behavior. Fig. 10 also shows predictions with varying stress levels. These predictions agree with trends observed and reported in the literature [16,37]. What this figure does not show is the dependence of the modification on the loading cycle.

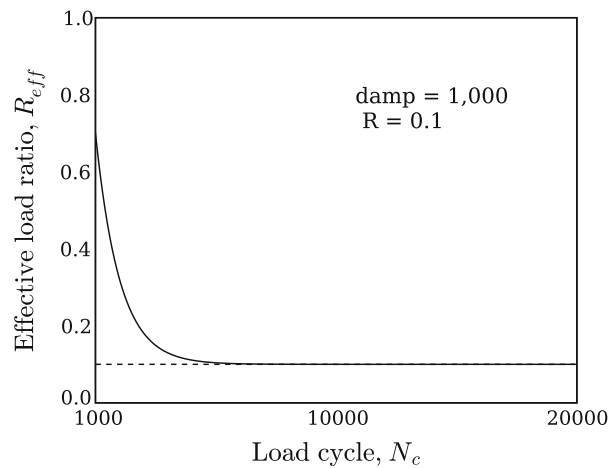


Fig. 8. Example of R_{eff} as a function of load cycle for $R = 0.1$.

Table 1
Crack growth parameters.

$damp$	1000	n'	3.5	p	0.5
K_{Ic}	1320 MPa $\sqrt{\text{mm}}$	n	2.80	q	1.0
K_{Ic}	972.7 MPa $\sqrt{\text{mm}}$	C	3.0e-8	A_k	0.75
ΔK_1	24.32 MPa $\sqrt{\text{mm}}$	C_m^{th}	0.1	B_k	2.0
R_{so}	3.0	C_p^{th}	1.3	a_0	0.00381 (mm)
α	2.0f	$\frac{S_{max}}{\sigma_0}$	0.3		

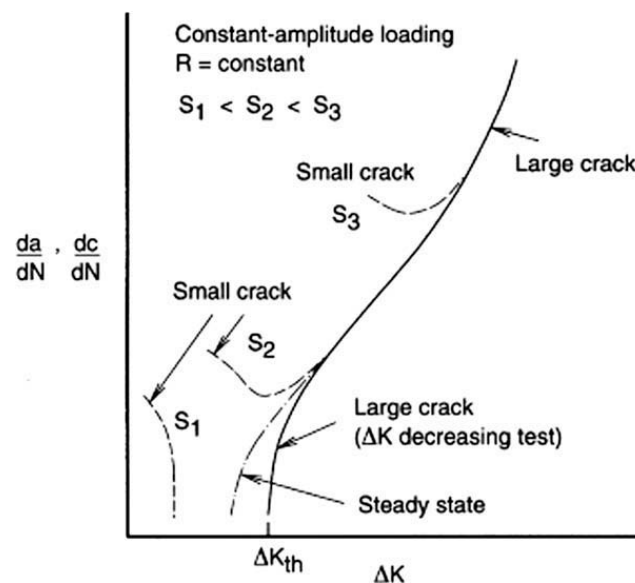


Fig. 9. Accelerated growth rates expected for young cracks at increasingly higher applied stress, taken from Newman et al. [36].

There is not enough data from the test to support the inherent time-scale introduced into the modified rate equation; however, the model performs as desired by predicting accelerated growth for young cracks which, without the modification, would be in the sub-threshold regime.

4.2.3. Load interaction effects

It has been observed that crack growth rate can dramatically decrease following an overload in an otherwise constant amplitude loading program. Conversely, following under-loads in the loading program crack growth rates accelerate; however, this effect is typically less pronounced. These load interaction effects are observed to be transient after which the growth rate returns to that observed under the constant amplitude loading [40]. One key feature of DDSim is the ability to treat variable amplitude loading spectra; consequently, a model for load interaction is essential.

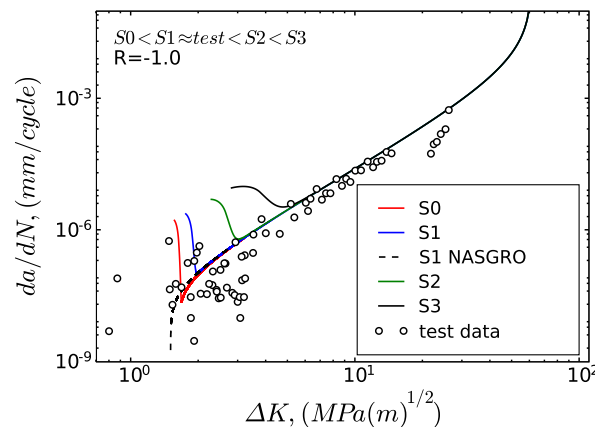


Fig. 10. Predicted fatigue crack growth rate of small cracks near threshold ΔK with increasingly higher applied stress and crack growth rate data [17].

Plasticity is responsible for load interaction [41]. Willenborg et al. [42] developed a model to account for load interaction that computes an effective stress range based on plastic zone size. Gallagher [43] reformulated this idea in terms of the stress intensity factor in the Generalized Willenborg model. With this model, the net result is a lower R for the same stress intensity range. Thus, the Generalized Willenborg model works in a similar manner to the young crack model and is implemented to treat load interaction in Level I. Although it is not generally conservative, the current implementation of Level I neglects accelerated growth due to under-loads because the spectrum of current focus contains few, low magnitude under-loads.

5. Procedure for low fidelity life prediction

Fig. 11 shows a flow chart describing DDSim Level I. The following subsections describe each component in the flowchart starting with the required input, then moving to the inner state-of-practice loop and working outward. Finally, a subsection summarizes the data flow and a concluding section discusses how the algorithm is modified for high performance parallel computing.

5.1. Input

The user input is the finite element model with its boundary conditions and computed fields, material parameters, and initial crack sizes.

5.1.1. Finite element model and material parameters

DDSim Level I is designed to operate on a user-supplied, finite-element-generated, elastic stress field. A method to read, to store and to query the finite element mesh and corresponding data was developed. The required inputs from the finite element analysis are the mesh and nodal stress values and both are provided in ASCII text files. If spectrum loading is to be considered, the spectrum is also required input. Otherwise, for constant amplitude loading the load ratio, R , is required input. The stress field could be generated for any applied loading case and, in the case of spectrum loading, a load factor is used to account for other load levels. For example, the finite-element-generated stress field could be computed for the maximum loading in the spectrum in which case the spectrum would be input as the fraction of the maximum load in the program.

Additional user input is required to perform the fatigue life calculations. This input is: initial flaw size, control crack growth increment, maximum practical life span (in loading cycles), the critical stress intensity factor, K_{Ic} , and the various material parameters in support of Eq. 15. The control crack growth increment, Δa_{crit} and the maximum practical life span, N_{max} , are structure/boundary condition specific and are provided as user control parameters. Both user input parameters are discussed below.

5.1.2. Initial crack sizes

DDSim Level I is designed to process either a list of deterministic initial flaw sizes, specified as the major and minor axis of an elliptical flaw and corresponding with the intrinsic material defect or the smallest flaw detectable by NDE, or a set of initial flaw sizes determined from a random sampling.

There are two methods of generating random initial flaw sizes implemented in Level I. The first is by sampling a three-parameter Weibull distribution. The probability density for a three-parameter Weibull distribution is given by

$$f_X(x) = \frac{\rho}{x_0} \left(\frac{x - \gamma}{x_0} \right)^{\rho-1} e^{-\left(\frac{x-\gamma}{x_0} \right)^\rho} \quad (16)$$

where γ is a truncation value, ρ is the Weibull shape parameter and x_0 is the scale parameter. The three-parameter Weibull distribution could be used to make the initial flaw sizes reflect distributions of observed metallurgical artifacts known to

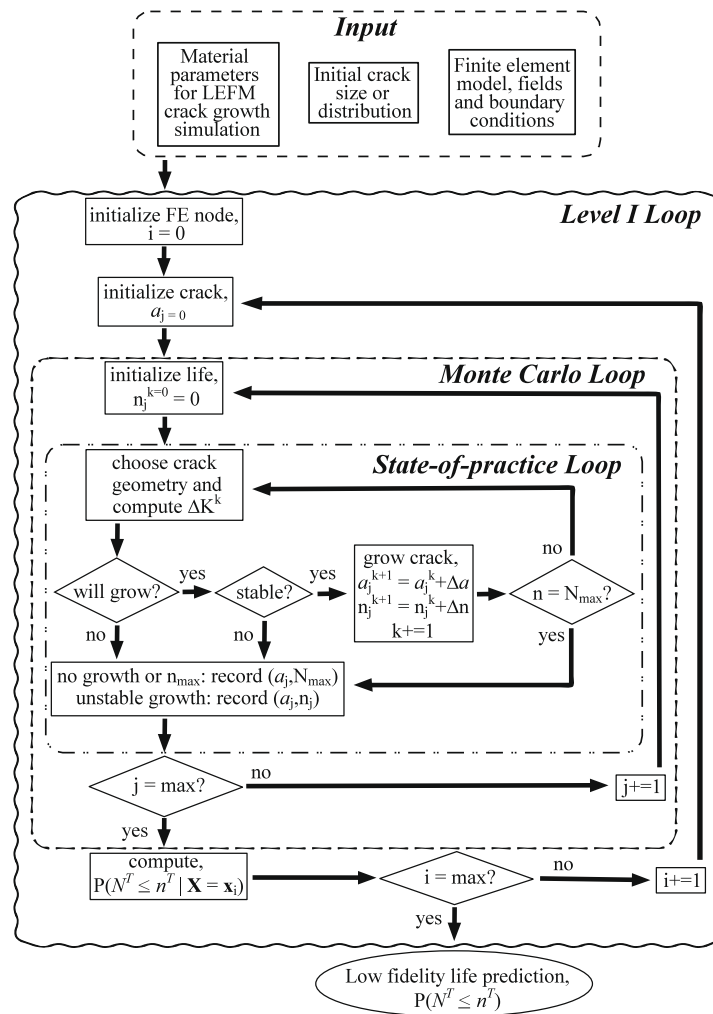


Fig. 11. Flowchart for the Level I procedure.

nucleate cracks, such as second-phase inclusions. The current implementation uses a distribution of second-phase particle area, obtained from observations, to determine the initial radius of a penny-shaped flaw.

The second method of generating random initial flaw sizes is yet more physically justifiable. Campman [13] found that for 7075-T651 fatigue cracks originate at broken second-phase constituents. Bozek et al. [44] developed a method to determine whether a given instance of a second-phase particle would crack. Bozek's method idealizes the particle as an ellipsoid. Then, it uses information about the particle shape and size, the lattice orientation of the grain surrounding the particle, and the local strain level in the surrounding grain to determine if the particle would break. Hence, given a statistically accurate set of constituent particles for a sample volume of 7075-T651, Bozek's method returns the subset of cracked particles.

From this subset, Level I uses the *maximum* dimension of each broken particle, in the plane normal to the material rolling direction, as the radius of an initial flaw. Not only does this provide a more physically justifiable set of initial flaws, it also provides information about the probability of crack incubation. This important observation will be repeatedly addressed throughout this section.

For 7075-T651, cracked particles are often located within plastically-deformed grains, an effect that Bozek's method incorporates. However, in Level I the finite-element-generated input stress field is intended to be linear elastic. This contradiction could lead to non-conservative results if the local strain was naïvely computed using an elastic Hooke's law. However, a more conservative estimate of the local strain can be computed from the elastic stress field if the strain is back-calculated using a deformation plasticity rule. Hence, Level I uses the Ramberg–Osgood deformation plasticity model to back calculate the local strains. It is not possible to say whether this provides any type of bound on the amount of plasticity. Without higher fidelity models, predictions on the amount of load shedding when the material begins to flow are not possible.

5.2. State-of-practice loop

The function of the innermost loop in Fig. 11 is to perform fatigue life prediction for a specified initial flaw at a specified damage origin. Doing so requires choosing the most appropriate SIF solution, computing the driving force, determining the

type of growth, computing the crack growth increment, changing the crack geometry and repeating as necessary. In the following discussion, the “choose crack geometry and compute ΔK^k ” box in the flowchart is broken into two subsubsections. The remainder of the operations are discussed in the subsubsection “Crack growth”.

5.2.1. Choosing the crack geometry

To choose the initial crack geometry, the elliptical flaw is assumed to originate perpendicular to the maximum principal stress at the damage origin. Hence, the x -axis of the ellipse, Fig. 12, is chosen to align with the first principal stress direction. Then the major axis of the crack is assumed to fall in the direction of the second principal stress direction and, of course, the z -axis corresponds to the third principal stress direction.

For cracks that fall on surfaces with non-zero traction boundary conditions or with applied displacement boundary conditions, it is possible that the first principal stress direction aligns with the surface normal. This pathology is illustrated in Fig. 12 (a) where the principal stress, σ_1 , is in the same direction as the surface normal, $\mathbf{n}$. Clearly, there is no interest in conducting a fatigue life prediction for a crack in this orientation. When this pathology is detected, the orientation of the flaw is permuted such that the x -axis is aligned with the second principal stress direction and the y -axis is aligned with the third principal stress direction, as shown in Fig. 12(b).

A second pathological crack orientation could arise when the crack origin is located at a geometric corner, as illustrated in Fig. 13. In Fig. 13(a), the default orientation of the crack falls parallel with one (or more) of the element surfaces in the mesh (not shown). In this case, the resolution is more complicated than choosing the second principal stress direction because there is no guarantee it will fall inside the volume. Consequently, the z -axis of the flaw is first chosen in the opposite direction of the average surface normal, $\mathbf{n}$. Then the y -axis of the flaw is given by the direction obtained from the cross product of the z -axis with the first principal stress direction. The x -axis is determined by the cross product between the y - and z -axes. Finally, the y - and z -axes are rotated 45° about the x -axis to land on the surface of the geometry. Note that for non-perpendicular corners, this final step leads to a possible source of error. However, an analytical solution for arbitrary corner angles does not exist, so this is the best approximation that can be made currently. The result of this process is shown in Fig. 13(b).

During the crack orientation process, the most appropriate crack shape for the geometry is chosen from information gathered during the orientation process. Fig. 14 illustrates three possible geometry cases. In case (a), the crack is located in a sliver. There is no appropriate stress intensity factor solution coded for this geometry, so the fatigue life for this case is zero. The geometry that leads to this case is not common in engineering structures where narrow, pointed edges are typically avoided. It could arise on bolt threads, but otherwise may be indicative of a modeling problem or error in the background finite element mesh. For case (b), for an internal angle θ between 15° and 125° , the quarter-elliptical stress intensity factor solution is used. For case (c), an internal angle θ between 125° and 345° , the semi-elliptical stress intensity factor solution is used. The full ellipse stress intensity factor solution is used for all the remaining possible geometries.

5.2.2. Computing the stress intensity range, ΔK

To compute the stress intensity factor, Level I uses the analytical stress intensity solutions described in Section 4.1 in combination with the principle of superposition. Level I samples the stress field over the crack surface, rotates it into a basis perpendicular to the crack surface and performs a least-squares fit to the degree polynomial required by the SIF solution. Then, the SIF is computed using the appropriate stress intensity factor solution. For the semi- and quarter-elliptical SIF solutions, the tension and bending components of stress are deduced from a linear least-squares fit in the local z -direction. Because everything is linear elastic, the appropriate loading factor is applied directly to the stress intensity and the stress intensity range is computed using the load ratio R for the given cycle.

5.2.3. Crack growth

After computing the stress intensity range for the initial state, an assessment is made as to whether the crack will grow. Accordingly, the SIF range is compared to the threshold range. No growth is predicted when the SIF range is less than threshold. One can note that due to the young crack modification in the crack growth rate (Eq. 14), every crack with positive SIF range grows for at least one cycle. In the event that crack growth fully arrests, the crack size is recorded with the inputted

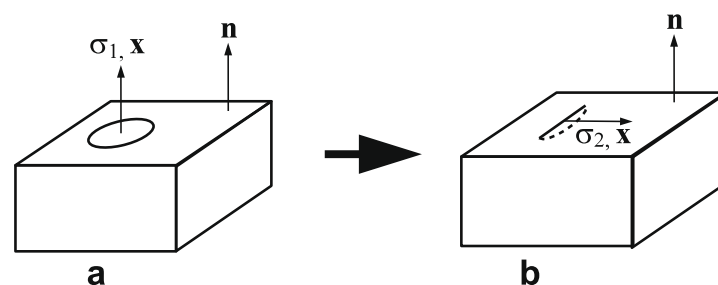


Fig. 12. Surface flaw orientation showing (a) a pathological default orientation and (b) the resolution.

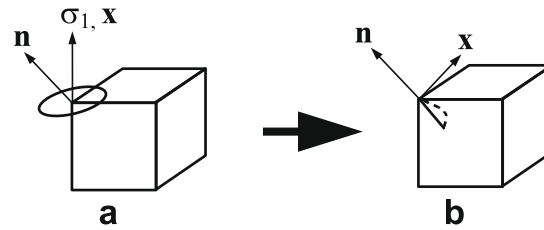


Fig. 13. Corner flaw orientation showing (a) a pathological default orientation and (b) the resolution.

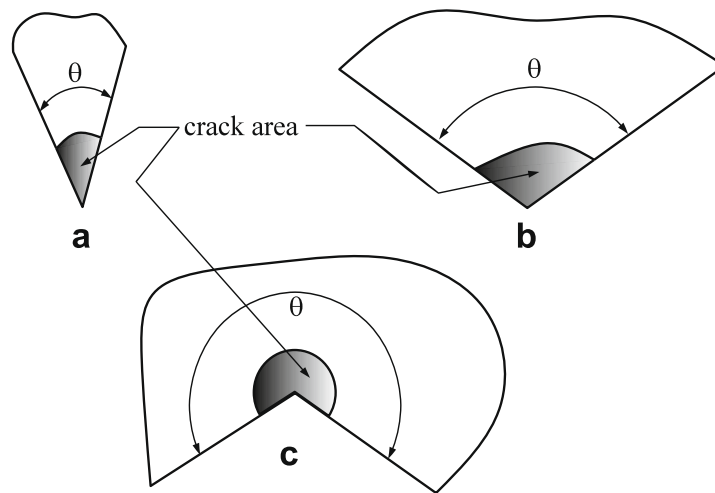


Fig. 14. Illustration of three possible geometry cases: (a) $\theta \leq 15^\circ$, (b) $15^\circ < \theta \leq 125^\circ$, and (c) $125^\circ < \theta \leq 345^\circ$.

specified maximum number of load cycles, N_{max} . Under spectrum loading, each distinct cycle in the spectrum is checked at least once before exiting the state-of-practice loop.

When the SIF range exceeds threshold, crack propagation is predicted. Unstable growth is predicted when one of two events occurs. The first of these is if the maximum stress intensity exceeds the material toughness. In Level I, the plane strain fracture toughness is used because it is conservative and because plate thickness is not readily ascertained to compute another value of critical stress intensity. The second event is if the crack outgrows its geometry (i.e. becomes a through crack). In this case, the current crack size is recorded with the number of cycles consumed to get to that crack size.

If the maximum stress intensity is less than the material toughness and the SIF range is greater than threshold, the crack will grow stably. For constant amplitude loading the crack growth increment is computed using an adaptive Runge–Kutta method. In this case, the control crack growth increment, Δa_{crit} , is used as the minimum increment. This parameter is controlled by the user with an input percentage of change in crack length. For spectrum loading, the crack growth increment is computed on a cycle-by-cycle basis and the Generalized Willenborg [43] crack retardation model is applied. In either case, the appropriate updates are made to the life and crack dimensions.

In the event that the crack grows stably until the practical maximum life is reached, the state-of-practice loop is exited and the current crack size is recorded with N_{max} . Otherwise, the crack growth loop is repeated. Fig. 15 demonstrates three possible scenarios that could arise during stable crack growth. In case (a), a crack originates within the volume and grows tangent to one boundary. When this scenario is detected, the crack immediately jumps to the semi-elliptical crack geometry. In case (b), the crack originates within the volume, but grows tangent to two surfaces making the quarter-elliptical crack geometry the most appropriate. Finally, in case (c), when a semi-elliptical surface crack grows tangent to another boundary, the crack is transitioned to the quarter-elliptical geometry. It is noteworthy that, although the geometry shown in Fig. 15 is nicely orthogonal, geometry that arises in practice may not necessarily be so ideal. Consequently, the logic to predict the best transition path is complicated and should be expected to have practical limitations.

If the stress intensity and crack growth rate are computed each cycle in the cycle-by-cycle crack growth for spectrum loading, the simulation processing time could become exorbitant. However, the difference in cycle-by-cycle crack growth increment would be small, especially when the maximum stress intensity is well below the material toughness. Hence, the change in crack geometry would be small and, consequently, the change in stress intensity factor range would be proportional to the change in loading from one cycle to the next. In this case, the stress intensity does not need to be recomputed on each cycle. Consequently, Level I only recomputes the stress intensity when the change in crack length exceeds some amount specified by Δa_{crit} . This is specified by the user via a critical percentage of change in crack length. The default value is 10%. In other words, by default, for cycle-by-cycle crack growth the SIF is only recomputed when the change in crack

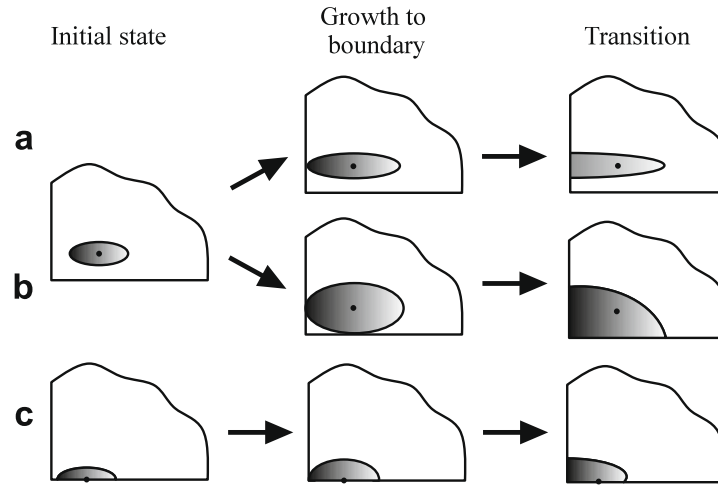


Fig. 15. Crack transition cases.

length exceeds 10% of crack size for which the SIF was last computed. The sensitivity to this parameter needs to be investigated by the user. For the validation problem in this paper, the life prediction at the most critical location was 7.8% higher for $\Delta a_{crit} = 10\%$ than for $\Delta a_{crit} = 0\%$. This is not conservative, but considered reasonable given the current scope.

5.3. Monte Carlo and Level I loops

For probabilistic life assessment, Level I performs Monte Carlo simulation at each damage origin using the damage tolerance procedure outlined above for each initial crack radius. The Monte Carlo loop initializes life, $n_i = 0$, and steps through the list of initial crack sizes at the same damage origin. Assuming the initial crack radii are less than a^{MLC} , the result at one node is a list of life predictions that corresponds to the list of initial crack sizes, Eq. 5. The list $\mathbf{n}_i^T$ represents a set of samples from the conditional distribution of total fatigue life given the event that a crack originates at node i

$$F_{N^T|\mathbf{X}}(n^T) = P[N^T(\mathbf{X}, A) \leq n^T | \mathbf{X} = \mathbf{x}_i] \quad (17)$$

where $A \in \mathfrak{R}$ is the random initial crack radius, and $\mathbf{x}_i \in \mathfrak{R}^3$ is the position of node i .

The Level I loop is wrapped around the Monte Carlo loop and iterates through the damage origins. For convenience, the finite element mesh nodes are used as damage origins. The option to restrict the simulation to surface nodes is implemented and used in the example problem because life-limiting cracks originate at or near free surfaces [33,45].

Upon completion of the Level I loop, the conditional probabilities for each node are combined, via the theorem of total probability, to give the cumulative probability of failure of the component as

$$F_{N^T}(n^T) = \sum_{i=1}^m P[N^T(\mathbf{X}, A) \leq n^T | \mathbf{X} = \mathbf{x}_i] \cdot P(\mathbf{X} = \mathbf{x}_i) \quad (18)$$

where m is the total number of nodes in the FE model, and $P(\mathbf{X} = \mathbf{x}_i)$ is the probability that the flaw originates at node i .

The probability of flaw incubation is assumed to be uniform for initial radii generated with the Weibull distribution. With this assumption, the cumulative distribution of failure becomes

$$F_{N^T}(n^T) = \frac{1}{m} \sum_{i=1}^m P[N^T(\mathbf{X}, A) \leq n^T | \mathbf{X} = \mathbf{x}_i]. \quad (19)$$

On the other hand, for initial flaws generated with Bozek's particle cracking filter, the probability of flaw incubation is given by the number of broken particles at node i , q_i , divided by the total number of broken particles, B . So

$$B = \sum_{i=1}^m q_i, \quad (20)$$

$$F_{N^T}(n^T) = \sum_{i=1}^m P[N^T(\mathbf{X}, A) \leq n^T | \mathbf{X} = \mathbf{x}_i] \cdot \left(\frac{q_i}{B}\right). \quad (21)$$

This distinction is very important. Assuming a uniform distribution of flaw incubation places equal weight on the conditional probability of failure for each node. In other words, the conditional probability at a node near a high stress concentration would have equal weight, $1/m$, as a node in a low stress region in the total structural reliability; this is unrealistic. On

the other hand, with the particle filter, the weight of a given node in the total structural reliability is based on the micro-structurally informed probability of particle cracking, q_i/B .

Finally, the probability of failure at some critical number of load cycles, n_{cr}^T , is

$$P_f = F_{N^T}(n_{cr}^T) = P[N^T(\mathbf{X}, A) \leq n_{cr}^T] \quad (22)$$

and the reliability is computed as the probability of non-failure, or $R = P[N^T(\mathbf{X}, A) > n_{cr}^T] = 1 - P_f$.

The probabilistic Level I simulation involves l life predictions at m finite element nodes, there are $l \times m$ fatigue life predictions required. For reasonable FE meshes with a reasonable sample of initial crack radii, the number of fatigue life predictions can readily climb into the millions. This can be reduced by only considering surface nodes, but the computational demand is still very high. However, with the parallel processing implementation developed for this paper, large simulations are completed in relatively short amounts of time. The computational demand, the system used to accomplish these simulations, and analysis statistics are discussed in Section 6.

The result of the Level I simulation is a scalar field, the conditional probability, or mean life prediction, or any other meaningful statistic, can be easily color-contoured and plotted on the surface of the component for effective visualization of results. This facilitates post-processing and provides visual guide to regions that are particularly susceptible to fatigue damage.

5.4. Parallel implementation

The fatigue life simulation described in this section is also coded for parallel processing. This is necessary due to the high number of fatigue simulations required for Monte Carlo simulation. The parallel processing is easily accomplished because of the independent nature of the fatigue simulations at each damage origin (i.e. finite element node). Fig. 16 shows a flowchart of the parallel implementation.

In the parallel implementation, the input is read by the managing processor which first generates the initial crack sizes. Then, the managing processor queries the finite element model to obtain a list of the damage origins. These typically coincide with the surface nodes in the finite element mesh. The managing processor subdivides the list of damage origins equally among the processes in the parallel job, including itself. In Fig. 16, m is the total number of processes. After the subdivision of the damage origins, each process independently performs the fatigue life simulation outlined in Fig. 11 for its portion of the total damage origins. Upon completion, the data is sent back to the managing processor for post-processing and computation of the cumulative distribution of structural reliability.

Depending on the node numbering in the finite element mesh, dividing the list of damage origins sequentially could result in an imbalance in processing time. For example, if the finite element nodes on the surface are numbered sequentially, the nodes at a positive stress concentration around a hole might end-up assigned to the same processor. In that case, that process would be required to conduct a large number of fatigue simulations. In contrast, the nodes at a negative stress concentration might end-up assigned to another processor. At these nodes, the stress intensity range would be negative. Since the first check that DDSim Level I conducts is whether stress intensity range exceeds the threshold value, fatigue simulations conducted on this processor would end promptly and it would remain idle while the other processors conduct their fatigue simulations. To combat this scenario, the subdivision of the damage origins is done by random selection. Thus, the processing load is more equally balanced among the individual processes.

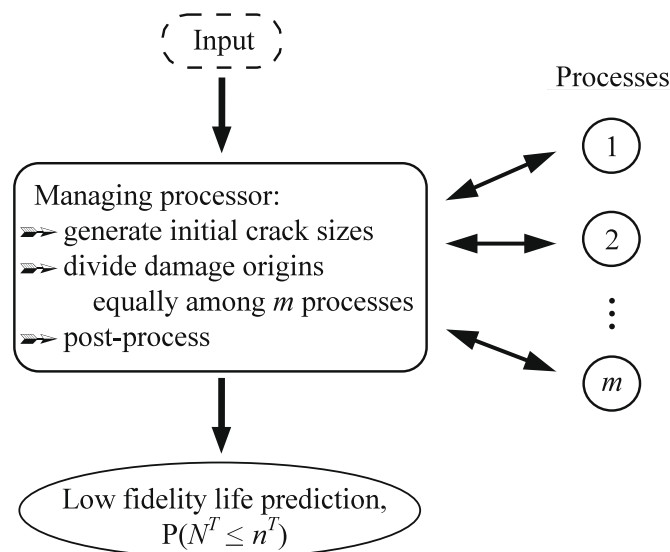


Fig. 16. The algorithm used to parallel process the Level I fatigue life simulations.

6. Validation case study: Level I analysis of a complex geometry, 7075-T651 laboratory test sub-component

In this case study, a complex geometry fatigue specimen is evaluated with DDSim Level I. This specimen geometry was tested in a controlled laboratory experiment five times by the Northrop Grumman Corporation (NGC), which provided the test data. The geometry of this specimen was chosen because it is representative of typical designs for structural wing-skin of aircraft. The specific dimensions of this geometry are not divulged herein because they are proprietary. However, sufficient information is provided to give the reader a clear sense of the geometry.

The purpose of this case study is to provide validating evidence for DDSim Level I. The following subsections describe the finite element model used as input, the material parameters used for crack growth, the initial crack sizes used in the Monte Carlo simulation, the Level I analysis results from a variable amplitude loading program, and a discussion of the results and comparison with test data.

6.1. FE model, material properties and stress analysis

There were two variants of the geometry used in the fatigue tests. Fig. 17 shows the model of the open-holed fatigue specimen. The specimen has six counter-sunk, counter-bored bolt holes at mid-length. Bolt holes 15 and 12 are present, but not labeled in Fig. 17(a). These bolt holes cause large stress concentrations, which result in the observed damage nucleation sites. The specimen is machined from rolled 7075-T651 plate. The specimen is modeled such that the x -direction corresponds to the material rolling direction (RD), the y -axis corresponds to the transverse direction (TD) and the z -axis corresponds with the normal direction (ND). The specimen is machined with thickened ends, required for the friction grips in the testing apparatus, and tapered down to the aircraft wing-skin thickness. There is a thickened pad at the row of bolt holes and the radii of the tapers from the gripped ends to the pad are designed to limit eccentricity in the load path as well as avoid artificial stress concentration. In the figures, there is an arrow indicating the forward and aft ends of the aircraft. This demarcation will be used for reference in the following results and discussion sections.

This geometry is *not* symmetric about the stiffener. There are two sources of asymmetry. First, the landing pad is thicker for holes #11–13 than for holes #14–16. Second, the space between hole 16 and the aft edge of the specimen is less than the space between hole 11 and the forward edge of the specimen. However, the center-to-center hole spacing is the same on both sides of the stiffener (see Fig. 18).

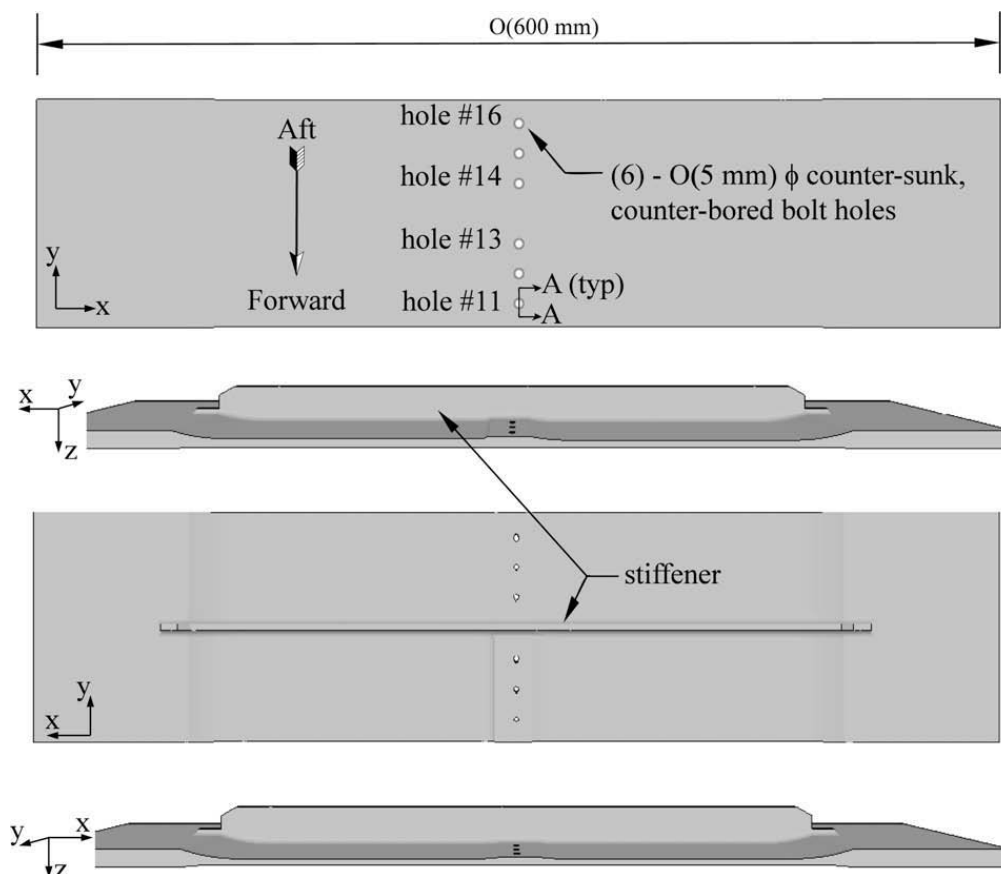


Fig. 17. The geometry of the open-holed fatigue specimen.

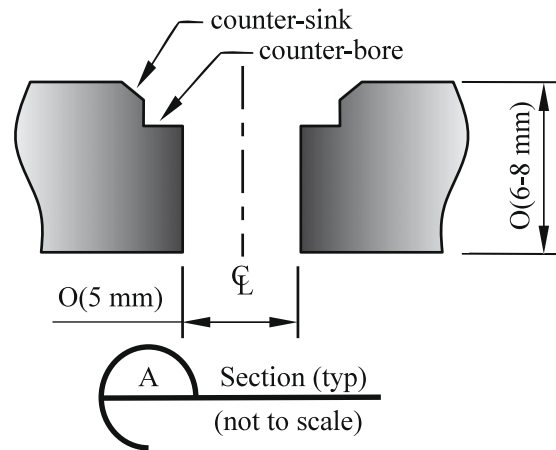


Fig. 18. Cross-section at the counter-sunk, counter-bored bolt hole.

Fig. 19 shows the model of the other fatigue specimen, with fasteners in the holes and clip-angles on the stiffener-side of the specimen. Again, bolt holes 15 and 12 are not labeled in Fig. 19(a). The fasteners and clip-angles are not included in the fatigue life simulation; however, they are present in the finite element model used to generate the stress field. The clips and fasteners are included to model the wing rib which reduces bending in the specimen and, consequently, the location of maximum stress around the bolt holes is repositioned. The stress field and boundary conditions are discussed in further detail below.

Fig. 20(a) shows the surface mesh on the model and (b) a close-up at one of the bolt holes. Fig. 20 (b) is typical for each hole. The same mesh is used for the models of both the open-holed specimen and the specimen with fasteners; however, interface elements were placed between the fasteners and clip-angles to model contact. In total, there are 140,024 8-noded brick elements and 184 6-noded wedge elements in the input to DDSim. There are 63,974 nodes on the surface of the model.

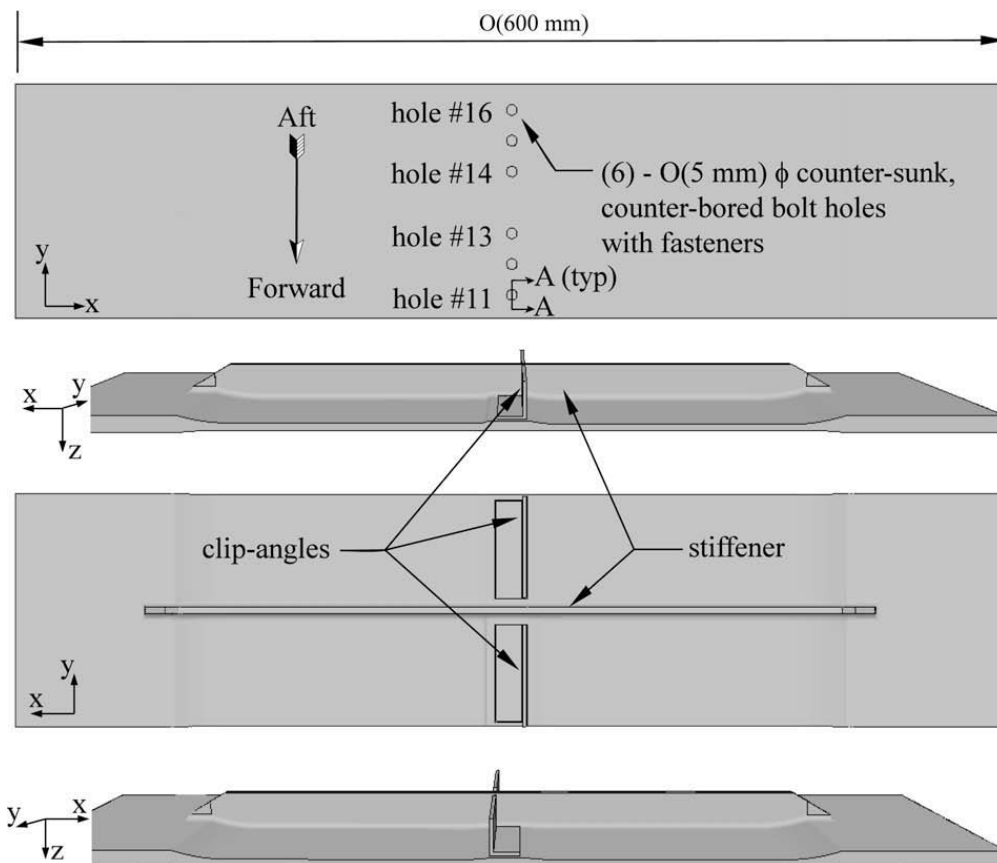


Fig. 19. The geometry of the fatigue specimen with fasteners and clip-angles.

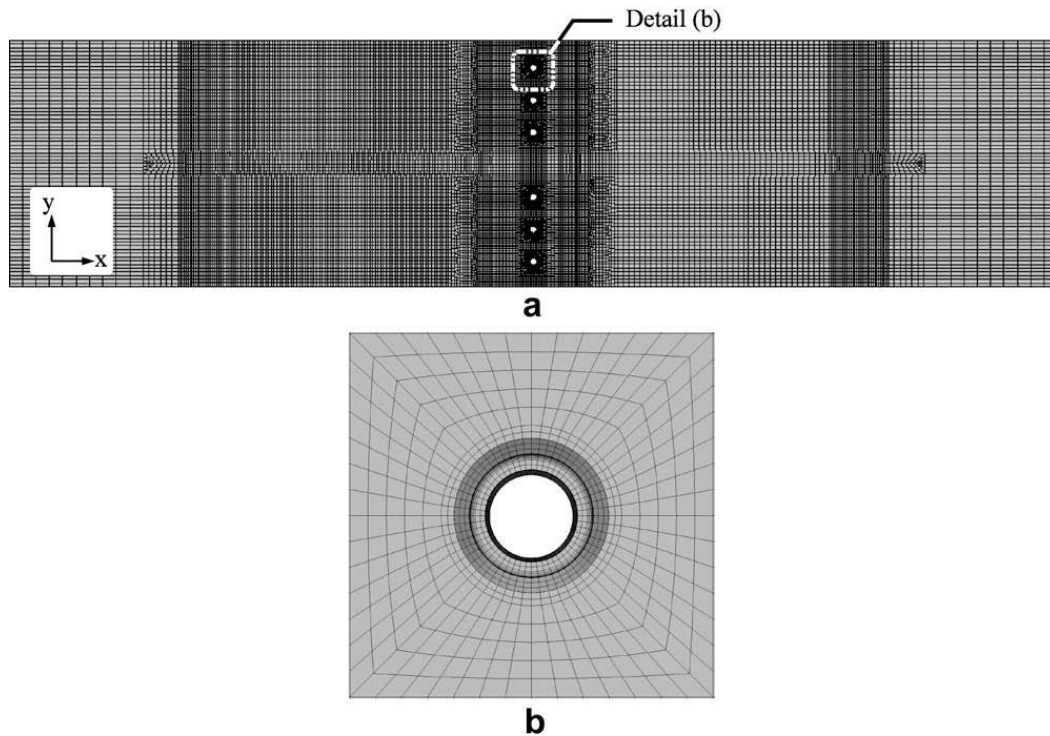


Fig. 20. The finite element mesh shown (a) globally and (b) locally near the bolt hole.

There was no mesh refinement study to explore the convergence of the computed fields performed for this validation. Mesh refinement would affect the life prediction in three ways. First, the accuracy of the stress fields would improve the life prediction. Second, there would be more surface nodes for initial flaw locations. However, for the low fidelity predictions of Level I, slightly more accurate stress fields and many more initial flaws would not be worth the increased cost. The third impact that mesh refinement would have is as cracks grow they would pass through the domain of more elements which would likely decrease the accuracy of the least-squares fit for the analytical SIF solutions.

Fig. 21 shows the boundary conditions used in the model. The displacements are labeled u , v , and w corresponding to the RD, TD and ND directions, respectively. The applied load for the variable amplitude simulation corresponds to the maximum load in the loading spectrum. The applied load for the constant amplitude simulation corresponds to the root mean square (RMS) of the spectrum. For the open-holed specimen, the maximum load in the spectrum is 316.8 kN and for the model with fasteners it is 378.1 kN. In the constant amplitude loading analysis, the RMS load is applied with full unloading, $R = 0$. For the open-holed specimen, the RMS load applied is 177.9 kN and for the model with fasteners it is 212.2 kN. Bolt pre-load is not considered in this example problem; however, it could be treated by DDSim if the stress analysis included bolt tension and the appropriate contact.

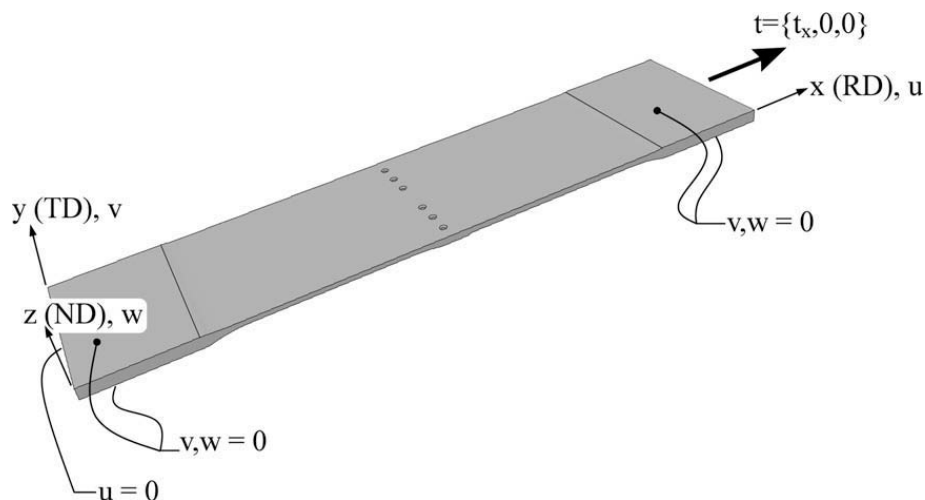


Fig. 21. The boundary conditions applied to both fatigue specimen models.

The constitutive model is linear elastic, isotropic. The modulus of elasticity, E , is 71 GPa, and the Poisson's ratio, ν , is 0.33.

Fig. 22 shows the first principal stress contour plot for (a) the open-holed specimen and (b) the specimen with fasteners subjected to the maximum load in the spectrum. For each hole in the open-holed specimen, the highest stress concentration occurs just below the intersection between the counter-bore and main-bore of the bolt hole. The magnitude of the stress concentration is smaller at the holes toward the middle of the specimen, holes 13 and 14, where the stiffener provides a load path. The highest stress concentration in the open-holed specimen is at the aft side of hole 16, indicated with an arrow in the figure. The lowest positive stress concentration in the open-holed specimen is at the aft side of hole 13. The highest stress concentration in the specimen with fasteners is not localized near the intersection between the counter-bore and main-bore. Rather, the concentration is uniform for the upper two-thirds to one-half of the main-bore. The highest stress concentration occurs at the forward side of hole 16, indicated by the arrow. The next highest concentration is at the aft side of hole 14. The pattern is reflected at holes 11 and 13 on the other side of the stiffener. The yy -component of stress helps explain these differences.

Fig. 23 shows the contour plot of the yy -component of stress for (a) the open-holed specimen and (b) the specimen with fasteners subjected to the maximum load in the spectrum. For all of the holes in the open-holed specimen the yy -component is zero at the forward and aft edges of the hole, which agrees with the traction free-boundary conditions. However, for the specimen with fasteners, at the aft side of hole 16 there are compressive stresses due to contact. Likewise, at the forward side of hole 14 there are compressive stresses due to contact. Again, the reflected pattern is witnessed in the specimen with fasteners at holes 11 and 13. These contact stresses are caused by the fasteners which attach the clip-angles.

6.2. Fatigue spectrum, crack growth material properties and initial flaws

The first 100 cycles of the fatigue spectrum used in this case study are shown in Fig. 24. The stress intensity axis has been normalized by that for the maximum load in the spectrum. There are a total of 5280 distinct load cycles in this program before the spectrum is repeated. The subsequent cycles follow a very similar pattern to the first 100 cycles. This spectrum was developed at NGC to model the service loads experienced by the aircraft. It can be seen that there are various overload and underload excursions in the first 100 cycles.

The crack growth material properties used to compute the crack growth rate, Eqs. 15 and 14, were already shown in Table 1. These parameters were taken from the NASGRO material database for 7075-T651 [7]. The N_{max} is set to 100,000 cycles and control parameter for Δa_{crit} is set to 10%.

Both methods of generating samples of initial flaw radii are used in this case study. For each method, a set of 10,000 initial flaw radii are generated. Fig. 25(a) shows the normalized histogram for the set of initial flaw radii generated from the three-parameter Weibull distribution. The parameters for the distribution are tuned to replicate the statistics for particle area in the TD–ND plane. The area is then converted to radius assuming a circular cross-section. The parameters used are $\gamma = 6.1 \mu\text{m}^2$, $\rho = 0.53$, and $x_0 = 18.5 \mu\text{m}^2$. These parameters are estimated from observations using the maximum likelihood estimator technique with a 99% confidence interval [46]. The solid curve in the figure shows the analytical probability density (refer to Eq. 16).

The normalized histogram of the set of particle semi-axis used with Bozek's particle cracking filter [44] is shown in Fig. 25(b). This histogram represents the largest semi-axis of an ellipsoidal particle in the TD–ND plane. The solid line in

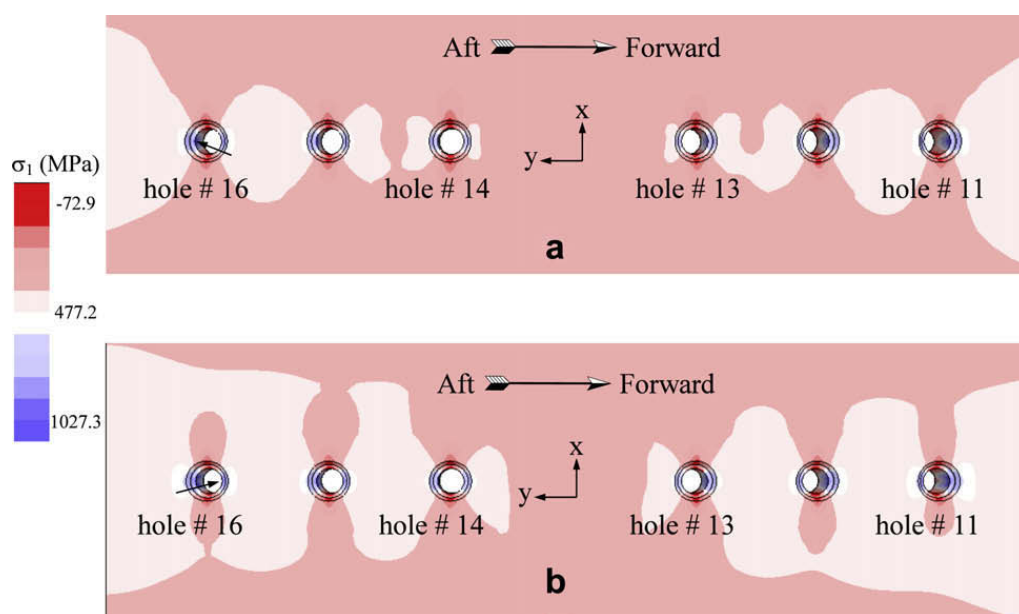


Fig. 22. The first principal stress contour plot for (a) the open-holed model and (b) the specimen with fasteners.

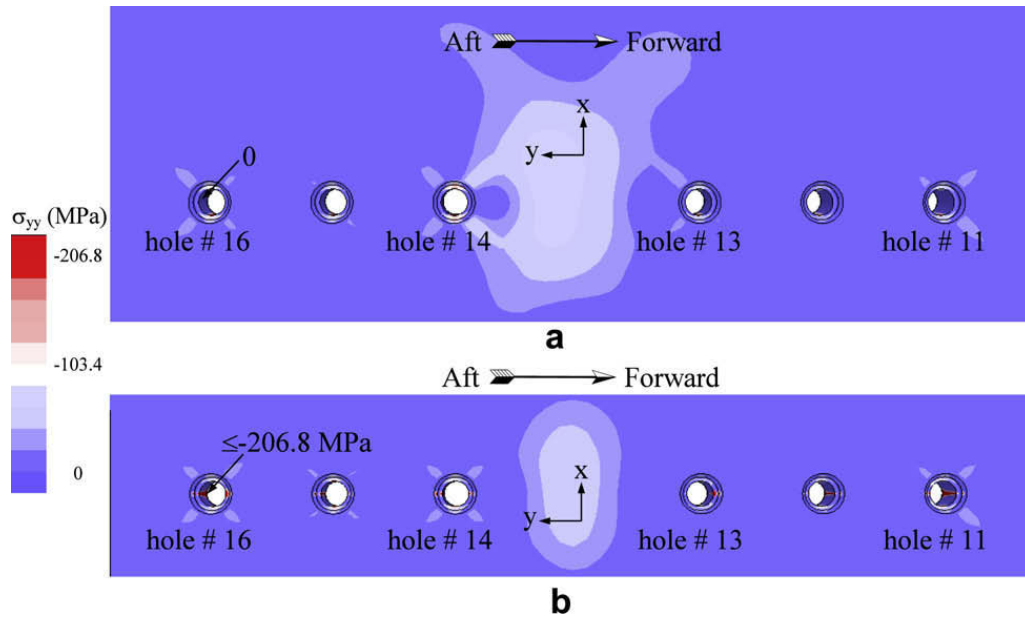


Fig. 23. The yy -component stress contour plot for (a) the open-holed model and (b) the specimen with fasteners. Note there is no σ_{yy} in the open-holed model.

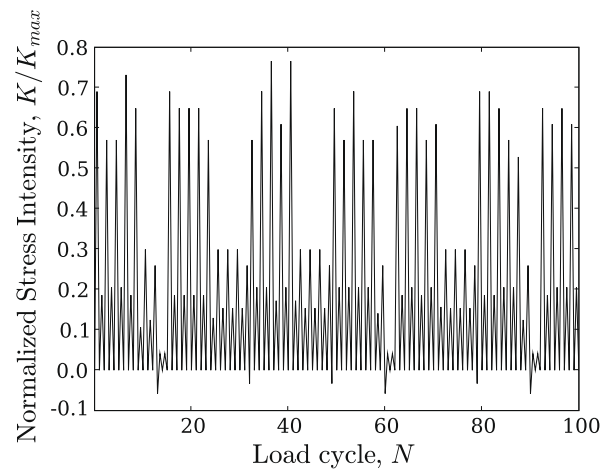


Fig. 24. The first 100 cycles in the fatigue spectrum normalized by the spectrum's maximum stress intensity factor.

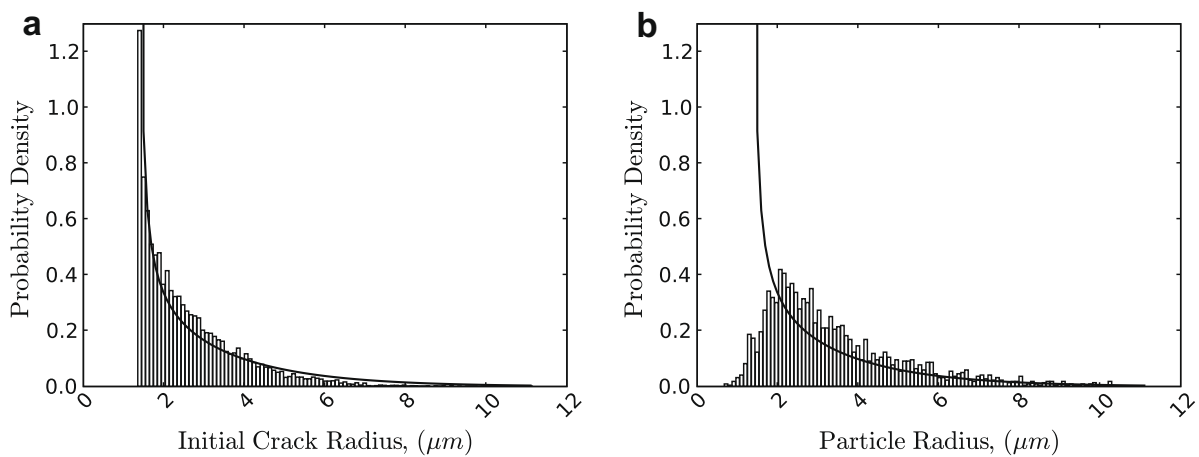


Fig. 25. Normalized histogram and probability density of initial flaw radii generated (a) with the three-parameter Weibull distribution ($\gamma = 6.1 \mu\text{m}^2$, $\rho = 0.53$, $x_0 = 18.5 \mu\text{m}^2$) and (b) from observed particle dimensions.

the figure is the analytical curve for the Weibull probability density used in Fig. 25(a). The obvious discrepancy is expected because the histogram represents the largest dimension of an elliptically shaped particle, whereas the analytical curve assumes circular particles. An important observation is that the set of initial flaws generated using the particle filter results in larger flaws and a larger quantity of large flaws.

6.3. Results

Fig. 26 shows the mean life prediction contour plot for the open-holed specimen under the variable amplitude loading program with initial flaw sizes from the particle cracking filter. The lowest mean life prediction is 29,058 cycles, which occurs at the aft side of hole 16 near the intersection of the counter-bore with the main-bore.

Fig. 27 shows the reliability of the open-holed model with initial flaws generated with the particle cracking filter and computed with Eq. 21. This figure includes the results from the constant amplitude and the variable amplitude loading programs. The curve for constant amplitude loading is the first to diverge from 100% reliability. Below approximately 60,000 cycles, the distribution for variable amplitude loading predicts a more reliable structure. Above 60,000 cycles, the predictions

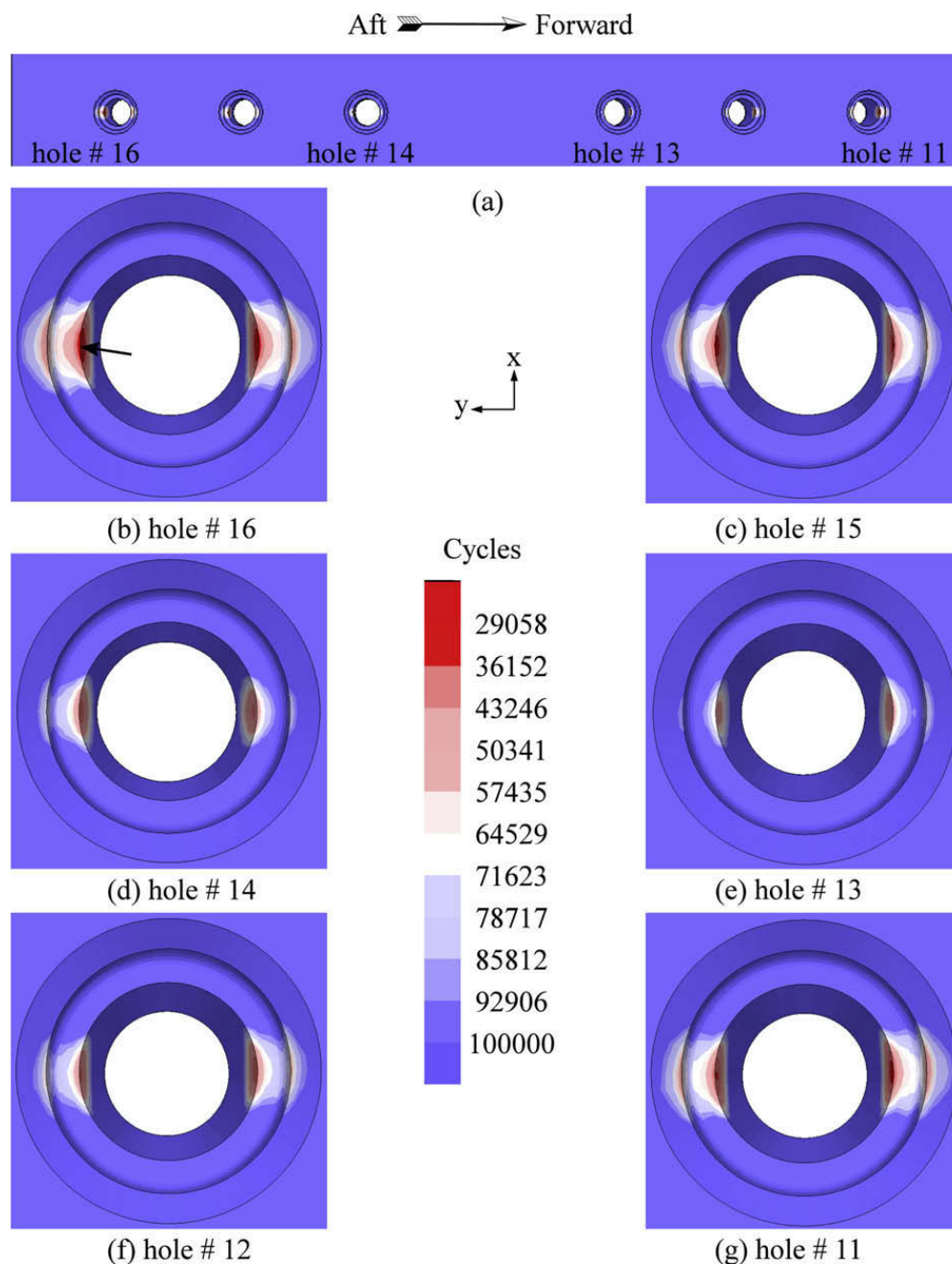


Fig. 26. The average life prediction for the open-holed model under variable amplitude loading with 10,000 initial particles. The minimum life occurs at the aft side of hole 16.

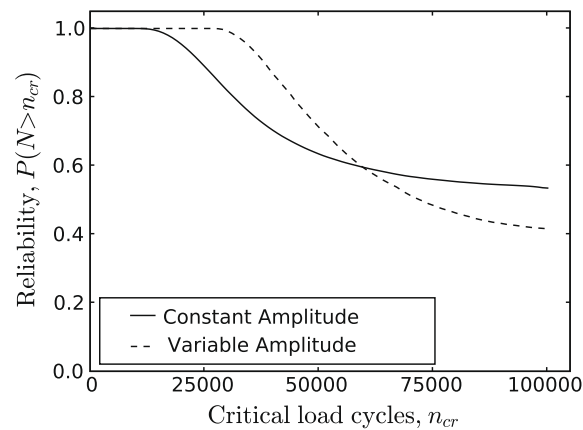


Fig. 27. Reliability for the open-holed model under both constant amplitude and variable amplitude loading with initial flaws generated with the particle cracking filter.

Table 2

Processing times, in minutes, for the parallel simulations of the open-holed model.

Processors	340		170		85	
	Average	Maximum	Average	Maximum	Average	Maximum
CA & Weibull	23.3	57.7	46.5	77.4	89.7	137.1
CA & particles	9.3	25.3	18.3	36.4	35.3	52.5
VA & Weibull	30.4	42.3	60.7	85.7	119.6	136.4
VA & particles	10.4	15.1	20.3	26.8	39.7	46.7

reverse. This observation suggests that spike loadings, present in the spectrum loading program, are more dangerous for larger cracks. Higher cycle count corresponds with larger crack size, where a spike in the load could cause unstable crack growth. Another participating factor in the difference between the shape of these curves is load interaction – constant amplitude loading has no load interaction. At lower cycle count, where unstable growth does not truncate life, load interaction retards crack growth in the variable amplitude simulation resulting in higher reliability.

The fatigue life simulations were run on parallel processors. For the validation problem, DDsim was run for four combinations of loading type and random initial crack size generation: constant amplitude with Weibull distribution, constant amplitude with the particle cracking filter, variable amplitude with Weibull distribution and variable amplitude with the particle cracking filter. Nearly linear scaling in the processing time is expected because of the decoupled nature of the simulation. Hence, a scaling study was conducted to determine how well the Level I code performed. The four combinations of open-holed simulations were run with 340–, 170–, and 85–3.6 GHz Intel Xeon processors with 2 GB of RAM. Table 2 summarizes the processing times of the simulations. The average processing time per processor and the maximum time for any processor are listed for each of the simulations. The table shows that average processor time scaled nearly linearly, as expected. The maximum processing time did not perform as well; however, considerable speed-up is still obtained. The maximum time represents the actual cost of the simulations.

Fig. 28 shows the mean life prediction contour plots for the specimen with fasteners and clip-angles under variable amplitude loading program with initial flaw sizes from the particle cracking filter. The lowest mean life prediction is 26,341 cycles and occurs at the forward side of hole 16. For the model with fasteners, the hot-spots move down to approximately two-thirds the height of the main-bore. The lowest mean life occurs under the constant amplitude loading program with initial flaws generated using the particle filter.

Fig. 29 shows the reliability of the specimen with initial flaws generated with the particle cracking filter. This figure includes the results from the constant amplitude and the variable amplitude loading programs. As is observed with the open-holed model, the curve for constant amplitude loading is the first to diverge from 100% reliability. Again, below approximately 60,000 cycles, the distribution for variable amplitude loading predicts a more reliable structure. Above 60,000 cycles, the predictions reverse. The reliability from the particle cracking filter has a much more realistic lower bound than does the reliability from the Weibull data.

The scaling study was also conducted for the model with fasteners and clip-angles. Similar performance was witnessed making it unnecessary to repeat the results here.

6.4. Discussion and comparison with experimental results

The variable amplitude simulations run with initial flaws generated from the particle cracking filter are considered to be the best choice for Level I. There are several reasons for this. First, using the variable amplitude loading capability is the most

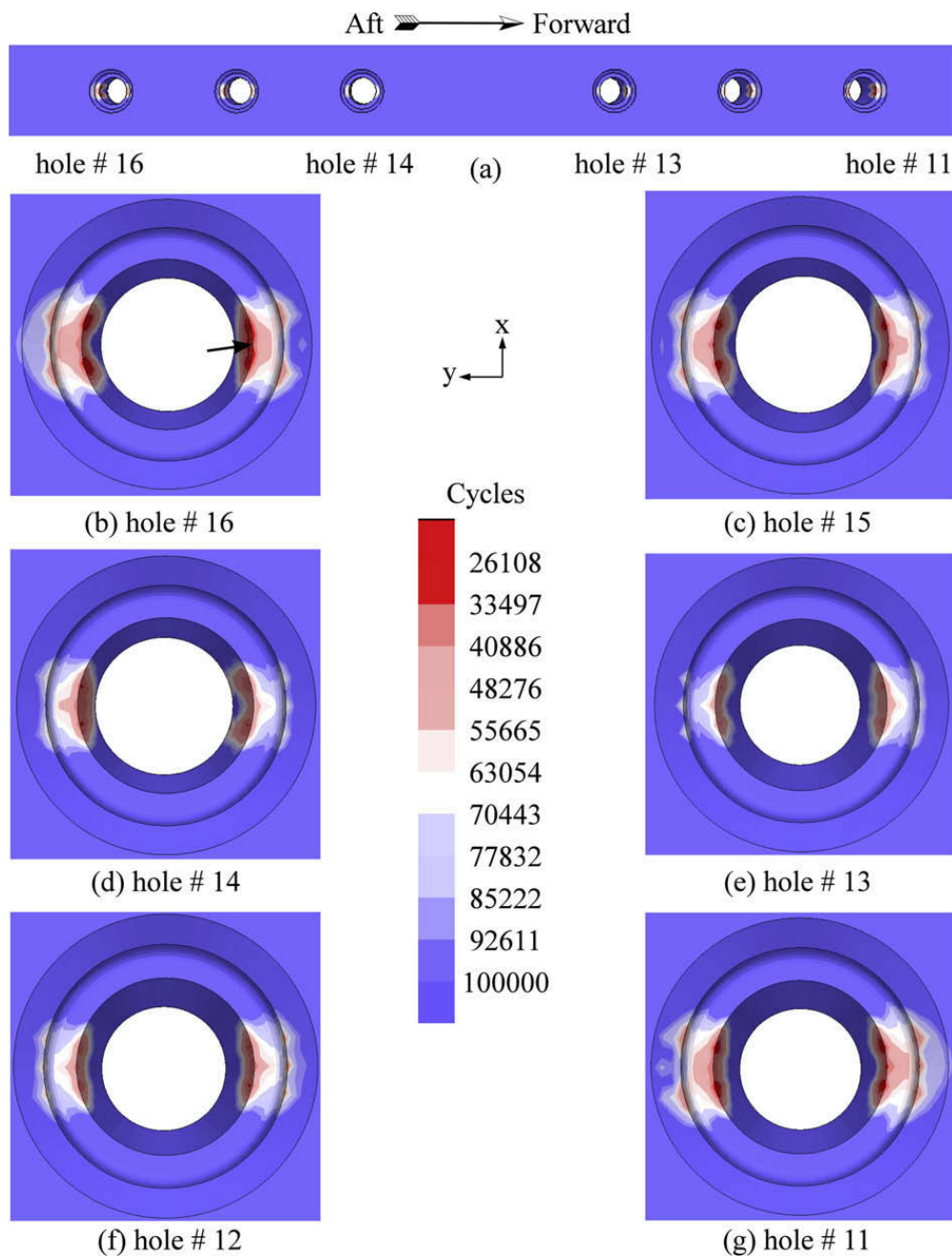


Fig. 28. The average life prediction for the model with fasteners under variable amplitude loading with 10,000 initial particles. The minimum life occurs at the forward side of hole 16.

accurate model of the actual boundary conditions in the laboratory. Second, the particle cracking filter gave physically justifiable initial flaw sizes derived from microstructural data. The three-parameter Weibull distribution, tuned to reflect observed particle area, is also microstructurally based. However, this method assumes the particles are circular in cross-section which is almost certainly not the case. Furthermore, with the Weibull distribution there is no physically justifiable method of determining the probability that a given node would nucleate a crack. Thus, it is assumed to be uniform; the undesirable results of which have already been discussed. Finally, the particle cracking filter is more likely to be conservative because it uses the larger dimension of the elliptical cross-section. Conservatism is desired because one of the tasks of the Level I simulation is to select sub-regions to perform higher fidelity analyses in Levels II and III. The following discussion, unless otherwise specified, will be based on the results from the variable amplitude simulations with the particle cracking filter. The other simulations will be mentioned as appropriate.

There were five specimens tested in the laboratory experiments conducted at NGC under the fatigue spectrum. One test was with open-holes and four included the fasteners and clip-angles. The open-holed specimen was loaded in fatigue until rupture. However, only one of the specimens with fasteners was run until rupture. Table 3 lists the experimental results. The first column gives the specimen identification number for reference. The second column identifies whether the specimen

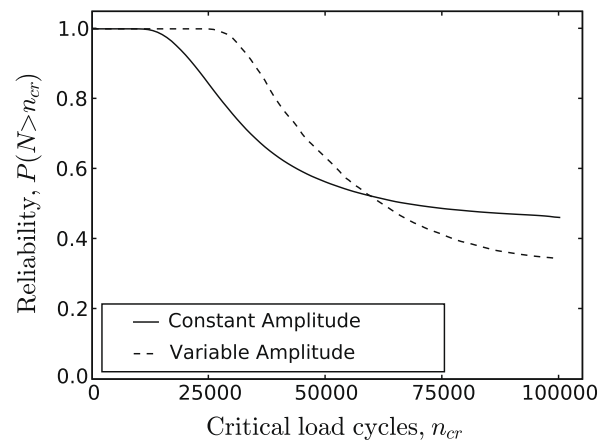


Fig. 29. Reliability for the model with fasteners under both constant amplitude and variable amplitude loading with initial flaws generated with the particle cracking filter.

Table 3

Experimental data used in validation case study.

ID #	Fasteners?	Maximum cycles	Rupture?
0773	No	53,485	Yes
0774	Yes	40,075	No
0775	Yes	21,418	No
0776	Yes	35,718	No
0777	Yes	30,998	Yes

included the fasteners and clip-angles. The third column reports the number of cycles of loading through the fatigue spectrum for the specimen. The final column indicates whether the test specimen was run to failure. The specimens that were not ruptured in fatigue were run in fatigue until a large crack was visually evident. Then, the specimen was monotonically loaded until the specimen ruptured.

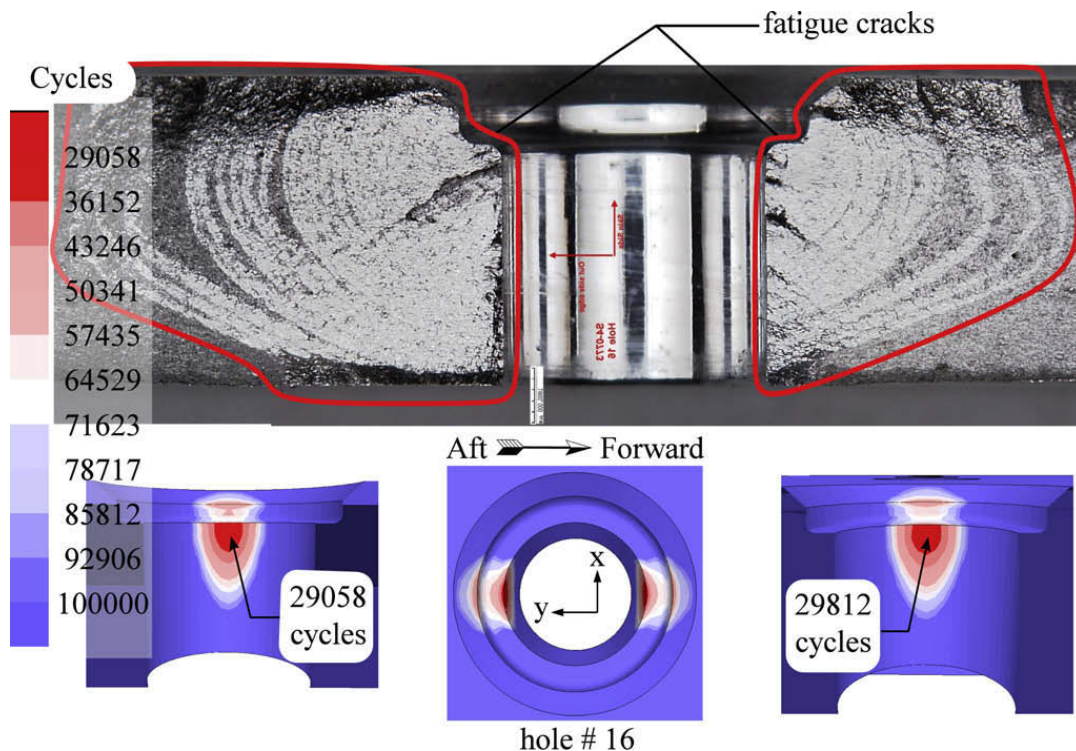


Fig. 30. Fracture surface at hole 16 compared with Level I simulation results for the open-holed model. Test #0773.

For specimen 0773, the specimen without fasteners, the number of cycles to failure was 53,485. The minimum average life prediction is 29,058 cycles, or 46% less than the test. Level I predicts the probability of failure at 53,485 cycles of 33.3%, or about 1 in 3. For specimen 0777, the specimen with fasteners that ruptured during fatigue loading, the number of cycles to failure was 30,998. The minimum average life prediction from Level I is 26,108 cycles, or 16% less than the test. The predicted probability of failure at 30,998 cycles is 4.0%, or about 1 in 25. From the limited statistics of specimens with fasteners loaded to, or above, 30,998 cycles, 1 in 3 failed, or 33%.

Through evaluation of the fracture surface, it was determined that the fatigue cracks of specimen 0774 were at, or very near, impending unstable crack growth. Considering impending instability to be failure, the number of cycles to failure for specimen 0774 were 40,075, or 26% greater than the minimum average prediction. Level I predicted the probability of failure at 40,075 cycles of 21.4%, or about 1 in 5. This small set of experimental data illustrates the stochastic nature of fatigue cracking.

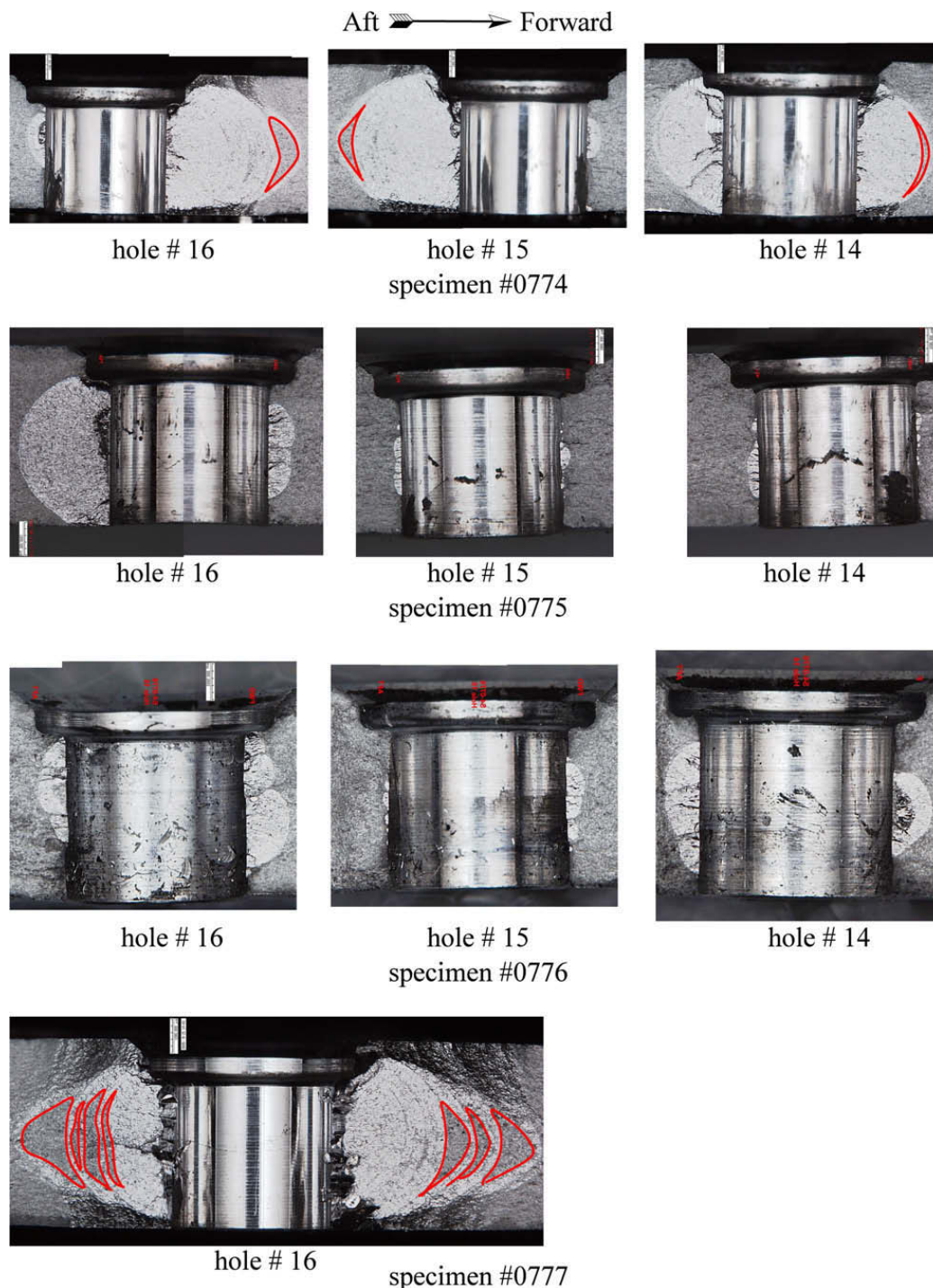


Fig. 31. The fracture surfaces for the specimens with fasteners and clip-angles.

Fig. 30 shows digital pictures of the fracture surfaces compared with Level I life prediction contour plots for hole 16 of the open-holed specimen. The life prediction contours include a view from the positive z-axis and cross-section views looking in the forward and aft directions. The fatigue crack surfaces are circled with a red line in the digital pictures. In reviewing the figures, it is important to keep in mind that the Level I simulations do not account for multiple crack interaction. This is not generally a conservative assumption and, in the sequel to this paper, the interaction effects are addressed with Level II analyses.

The fatigue surfaces in hole 16 showed large crack growth increments towards the outward edge of the fatigue surfaces. These increments are indicative of rapid growth and impending unstable crack growth. The fatigue crack in the aft side of hole 16 had the largest growth increments and the greatest surface area and, therefore, appeared to be the dominant crack. From these surfaces, it is evident that the origin of the fatigue cracks, in particular the dominant crack, is near the corner created by the intersection of the counter-bore with the main-bore. The Level I simulation matches this observation. For this geometry and loading, this location is near the location of maximum principal stress; however, this may not generally be the case because the life prediction accounts for the stress gradient, on the crack surface, as the crack moves away from the free surface.

Fig. 31 shows the digital pictures of the fracture surfaces of holes 14–16 for specimens 0774–0777. The fracture surfaces of specimen 0777 at holes 14 and 15 were marred by the fasteners upon rupture. Consequently, the digital images of those surfaces provided no information and the figure only includes a digital picture from hole 16. The digital pictures of holes 11–13 did not show evidence of large fatigue crack growth.

Fig. 32 shows the Level I predictions for the model with fasteners and clip-angles. The figure includes views looking forward and aft in each of holes 14, 15, and 16. The primary observation is that the life contours extend much further downward in the main-bore due to the clip-angles which reduce bending. Also, the hot-spots moved to the forward edge of hole 16. This is a result of the contact stress, normal to the surface in the y-direction, on the aft edge of hole 16 and forward edge of hole 14 caused by the fasteners and clip-angles.

In specimen 0774, the fatigue crack in the forward edge of hole 16, the aft edge of hole 15 and the forward edge of hole 14 each showed large and rapid increases in crack area toward the end of their fatigue growth. The large increases in crack area

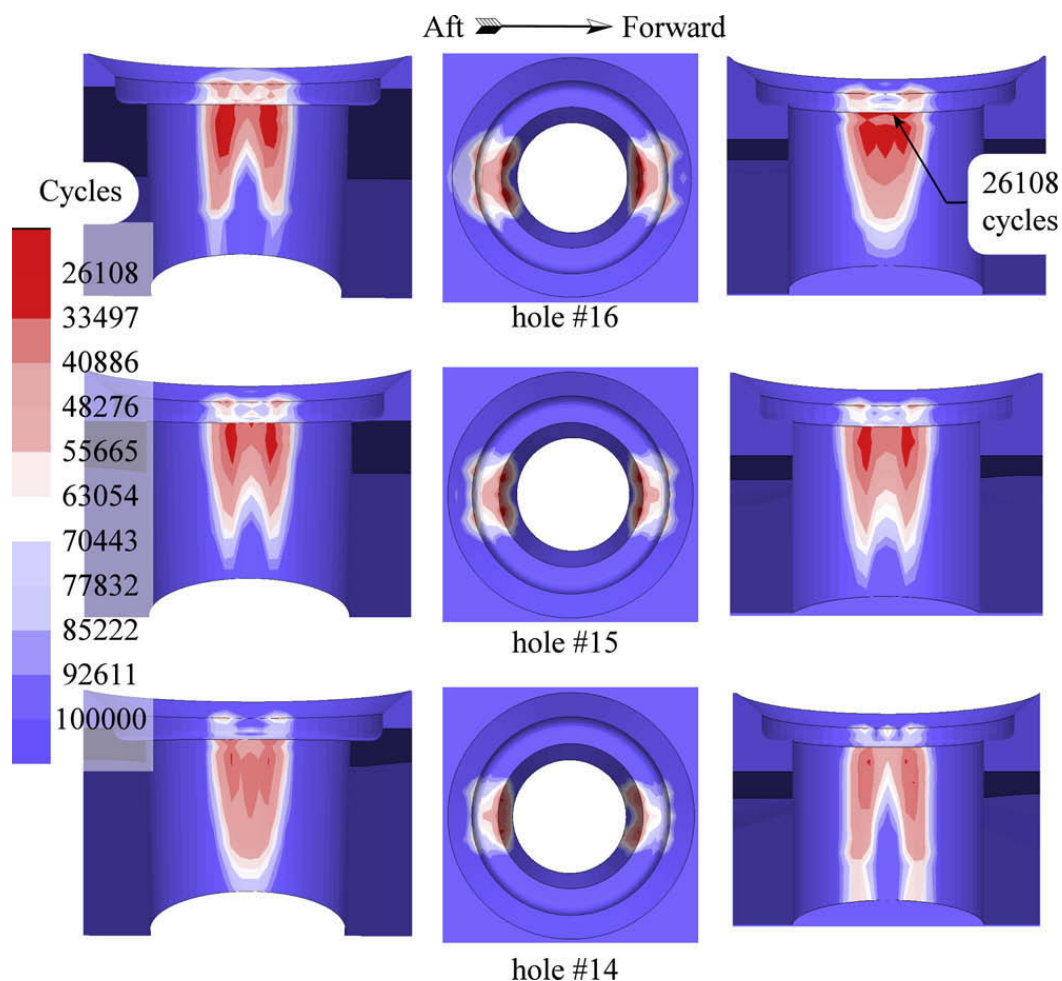


Fig. 32. The Level I mean life predictions showing forward and aft views.

are circled with a red line in Fig. 31. These large jumps in crack area are indicative of large, nearly unstable, steps of growth coinciding with peaks in the fatigue spectrum followed by stable fatigue growth during the post-peak loads. The fatigue crack at hole 16 in specimen 0777 showed similar jumps in crack area. Comparing the fracture surfaces of the two specimens, it is obvious that specimen 0774 was at, or very near, rupture during fatigue loading. Furthermore, the dominant crack in specimen 0774 appeared to originate at the forward edge, at approximately mid-height, in the main-bore of hole 16. This observation is consistent with the hot-spot predictions of Level I.

The fracture surfaces of specimen 0775 and 0776 did not show signs of impending instability. Specimen 0775 apparently contradicts the predicted hot-spot locations with the larger fatigue crack emanating from aft side of hole 16. This could have been due to a rapid increase in the incubation process caused by statistically unusual local metallurgical events, surface damage or environmentally assisted damage. In future work, some of these effects could be accounted for to guarantee conservative answers in Level I. Furthermore, they will need to be included in the higher-order analyses of Level III. The fatigue cracks in specimen 0776 appeared to agree well the Level I predictions showing large fatigue crack growth from the forward edge of hole 16 and the aft edge of hole 14.

7. Conclusions

DDsim Level I is the first hierarchical level in a multiscale fatigue life prediction methodology. It is designed to fulfill three tasks: select subdomains for the higher level simulations; produce a low fidelity life prediction to be updated by the higher level simulations; and produce a life prediction for preliminary design and scoping. Level I uses classical, linear elastic fracture mechanics in conjunction with analytical stress intensity factors, finite element generated stress fields, and empirical crack growth rate data to make its fatigue life predictions.

The crack growth rate equation used in Level I comes from the open literature, but is modified to model the accelerated growth rates observed with young cracks. The modification, developed in this work, is validated against experimental fatigue crack growth rate data and provides reasonable estimates. The modified rate equation requires a tuning parameter and a value of this parameter for 7075-T651 is recommended.

To obtain the probabilistic life prediction required in the new methodology, Level I uses Monte Carlo simulation. The Monte Carlo simulation is performed at each finite element node from the input finite element model and, consequently, can become very computationally intensive. Hence, a parallel version of Level I is developed to run on a parallel cluster of microprocessors. The nature of the simulation predisposes itself to parallel processing and nearly linear speed-up is achieved.

Finally, Level I is validated against five laboratory experiments performed at Northrop Grumman Corporation. There were two variants of the same aerospace structure tested. One variant had open bolt holes, while the other had fasteners and clip-angles. The geometry and boundary conditions of the experiments were modeled with a finite element model whose stress output is used as input to the Level I simulation. Level I simulations are conducted considering the root mean square of the applied fatigue spectrum and the applied spectrum with both Weibull generated initial flaws and initial flaws generated using the particle cracking criteria of Bozek et al. [44].

The validation showed that Level I gives the best results when the variable amplitude loading spectrum is used in conjunction with the particle cracking filter. The results from this validation show that the Level I reliability predictions are:

- (1) reasonably accurate considering the lack of fidelity included;
- (2) justifiably conservative; and
- (3) lead to appropriate hot-spot location selection.

Levels II and III were briefly described in the context of their rolls in the design of overall DDSim system. Detailed description of their implementation, verification, and validation will be presented in subsequent papers. Greater computing resources and improved computing technology will result in the capability to include increasingly high resolution in the multiple length scale models. However, there is still basic research that is required to fully exploit the computing resources of the future. The models used in Level III are still phenomenological. At these length scales, crystal plasticity is an improvement over classical J2-type plasticity; however, it is not sufficient to describe the multitude of mechanisms involved in microstructurally small crack growth. Many questions remain. For example, how should one adequately model grain boundaries, cell walls, persistent slip bands, and environmental effects? The authors believe that progress is contingent not only on technological advances in hardware and software, but with the mechanics and physics at these length scales.

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References

- [1] Constellation University Institute Project, http://exploration.grc.nasa.gov/Exploration/external/cuip_about.htm.
- [2] DARPA, DSO – Prognosis program, <http://www.darpa.mil/dso/thrust/matdev/prognosis/index.html>.
- [3] Farrar CR, Sohn H, Hemez FM, Anderson MC, Bement MT, Cornwell PJ, et al. Damage prognosis: current status and future needs, Tech. Rep. LA-14051-MS, Los Alamos National Laboratory; 2003.
- [4] Harris CE, Starnes Jr JH, Shuart MJ. Advanced durability and damage tolerance design and analysis methods for composite structures, Tech. Rep. NASA TM-2003-212420, National Aeronautics and Space Administration; June 2003.
- [5] Grandt AF. Introduction to damage tolerance analysis methodology. Damage tolerance of metallic structures: analysis methods and applications. Philadelphia, PA: American Society for Testing and Materials; 1984. p. 3–24.
- [6] Suresh S. Fatigue of materials. Cambridge University Press; 1998.
- [7] NASGRO fatigue crack growth analysis software, Version 4.2, Southwest Research Institute and National Aeronautics and Space Administration; 2004.
- [8] Harter JA. AFGROW users' guide and technical manual, Tech. Rep. AFRL-VA-WP-1999-3016, Air Force Research Laboratory, WPAFB, OH 45433-7542; 1999.
- [9] FRANC3D, Three-dimensional fracture analysis code, <http://www.cfg.cornell.edu/software/software.htm>, The Cornell Fracture Group, Cornell University, Ithaca, NY.
- [10] Millwater HR, Wu Y, Dias JB, McClung RC, Raveendra ST, Thacker BH. The NESSUS software system for probabilistic structural analysis. In: The 5th international conference on structural safety and reliability, San Francisco, CA; 1989.
- [11] Enright MP, Hudak SJ, McClung RC, Millwater HR. Probabilistic-based system for prognosis of fatigue in aircraft engine components. In: The 44th AIAA/ASME/ASCE/AHS structures, structural dynamics, and materials conference, Norfolk, VA; 2003.
- [12] Vasudevan AK, Sadananda K, Glinka G. Critical parameters for fatigue damage. Int J Fatigue 2001;23:S39–53.
- [13] Campman R. Statistical analysis of second phases particles in AA 7075-T651, Master's thesis, Carnegie Mellon University, Pittsburgh, PA; 2005.
- [14] Xue Y, Kadiri HE, Horstemeyer MF, Jordon JB, Weiland H. Micromechanisms of multistage fatigue crack growth in a high-strength aluminum alloy. Acta Mater 2007;55(6):1975–84.
- [15] Merati A, Eastaugh G. Determination of fatigue related discontinuity state of 7000 series of aerospace aluminum alloys. Engng Fail Anal 2007;14:673–85.
- [16] Lankford J. The growth of small fatigue cracks in 7075-T6 aluminum. Fatigue Engng Mater Struct 1982;5(3):233–48.
- [17] Newman Jr JC, Wu XR, Venneri SL, Li CG. Small-crack effects in high-strength aluminum alloys, Tech. Rep. NASA RP-1309, National Aeronautics and Space Administration; May 1994.
- [18] Iesulauro E. Decohesion of grain boundaries in three-dimensional statistical representations of aluminum polycrystals, PhD thesis, Cornell University; 2006.
- [19] Veilleux MG. Finite element model generation of statistically accurate 7075-T651 aluminum alloy microstructures, Master's thesis, Cornell University; May 2007.
- [20] Brahme A, Alvi MH, Saylor D, Fridy J, Rollett AD. 3D reconstruction of microstructure in a commercial purity aluminum. Scripta Mater 2006;55:75–80.
- [21] Saylor DM, Fridy J, El-Dasher BS, Jung KY, Rollett AD. Statistically representative three-dimensional microstructures based on orthogonal observation sections. Metall Mater Trans 2004;25A(7):1969–79.
- [22] Matous K, Maniatty A. Finite element formulation for modelling large deformations in elasto-viscoplastic polycrystals. Int J Numer Meth Engng 2004;60(14):2313–33.
- [23] Bozek JE. A 2D multiscale procedure for fatigue crack nucleation, Master's thesis, Cornell University; May 2007.
- [24] Roy A, Saha TK. Weight function for an elliptic crack in an infinite medium. I. Normal loading. Int J Fracture 2000;103:227–41.
- [25] Raju IS, Newman Jr JC. Stress-intensity factors for two symmetric corner cracks, fracture mechanics, ASTM STP-677, Philadelphia, PA: American Society for Testing and Materials; 1978. p. 411–30.
- [26] Raju IS, Newman Jr JC. Stress-intensity factors for a wide range of semi-elliptical surface cracks in finite-thickness plates. Engng Fract Mech 1979;11(4):817–29.
- [27] Newman Jr JC, Raju IS. Analysis of surface cracks in finite plates under tension or bending loads, Tech. Rep. NASA TP-1578, National Aeronautics and Space Administration; December 1979.
- [28] Newman Jr JC, Raju IS. Stress-intensity factor equations for cracks in three-dimensional finite bodies. In: Fracture mechanics, Vol. 1: theory and analysis, ASTM STP-791, vol. 1. Philadelphia, PA: American Society for Testing and Materials; 1981. p. 238–65.
- [29] Newman Jr JC, Raju IS. Stress-Intensity factor equations for cracks in three-dimensional finite bodies subjected to tension and bending loads. Computational methods in the mechanics of fracture. Elsevier Science Publishers, B.V.; 1986. p. 311–34.
- [30] Paris PC, Erdogan F. A critical analysis of crack propagation laws, transactions of the ASME. J Basic Engng 1963;85:526–34.
- [31] Donnelly E, Nelson D. A study of small crack growth in aluminum alloy 7075-T6. Int J Fatigue 2002;24:1175–89.
- [32] Lawson L, Chen EY, Meshii M. Near-threshold fatigue: a review. Int J Fatigue 1999;21:S15–34.
- [33] Newman Jr JC. The merging of fatigue and fracture mechanics concepts: a historical perspective, In: Fatigue and fracture mechanics, vol. 28, ASTM STP-1321, American Society for Testing and Materials, Philadelphia, PA; 1997. p. 3–51.
- [34] Newman Jr JC. A crack opening stress equation for fatigue crack growth. Int J Fracture 1984;24(4):R131–5.
- [35] Forman RG, Mettu SR. Behavior of surface and corner cracks subjected to tensile and bending loads in Ti–6Al–4V alloy. In: Fracture mechanics: 22nd symposium, ASTM STP-1131, vol. 1. Philadelphia, PA: American Society for Testing and Materials; 1992. p. 519–46.
- [36] Newman Jr JC, Phillips EP, Swain MH. Fatigue-life prediction using small-crack theory. Int J Fatigue 1999;21:109–19.
- [37] Newman Jr JC. A review of modeling small-crack behavior and fatigue-life predictions for aluminum alloys. Fatigue Fract Engng Mater Struct 1994;17(4):429–39.
- [38] Schijve J. Fatigue of structures and materials in the 20th century and the state of the art. Int J Fatigue 2003;25:679–702.
- [39] McEvily AJ, Ishihara S. On the dependence of the rate of fatigue crack growth on the $\sigma_a^m(2a)$ parameter. Int J Fatigue 2001;23:115–20.
- [40] Skorupa M. Load interaction effects during fatigue crack growth under variable amplitude loading – a literature review. Part I: empirical trends. Fatigue Fract Engng Mater Struct 1998;21:987–1006.
- [41] Anderson TL. Fracture mechanics: fundamentals and applications. CRC Press; 1995.
- [42] Willenborg J, Engle RM, Wood HA. A crack growth retardation model using an effective stress concept, Tech. Rep. AFFDL-TM-71-1-FBR, Air Force Flight Dynamics Laboratory, WPAFB, OH 45433; January 1971.
- [43] Gallagher JP. A generalized development of yield zone models., Tech. Rep. AFFDL-TM-74-28-FBR, Air Force Flight Dynamics Laboratory, WPAFB, OH 45433; January 1974.
- [44] Bozek JE, Hochhalter JD, Veilleux MG, Liu M, Heber G, Sintay SD, et al. A Wawrzynek and A R Ingraffea, A geometric approach to modeling microstructurally small fatigue crack formation: I. Probabilistic simulation of constituent particle cracking in AA 7075-T651, Modelling Simul Mater Sci Eng 2008;16:28. doi:10.1088/0965-0393/16/6/065007.
- [45] Zilberstein V, Schlicker D, Walrath K, Weiss V, Goldfine N. MWM eddy current sensors for monitoring of crack initiation and growth during fatigue tests and in service. Int J Fatigue 2001;23:S477–85.
- [46] Harlow DG. Private communication; 2006.